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KOD for H2 Compressor (175-D)

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\175-D 7%.pvdb

Date of Analysis : Sep 16, 2026 2:33pm

PV Elite 28 U1, March 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 506.000 mm. ID
Vessel Design Length, Tangent to Tangent 2503.20 mm.
Distance of Bottom Tangent above Grade 10000.00 mm.
Specified Datum Line Distance 0.00 mm.
Internal Design Temperature 130 °C
Internal Design Pressure 49.507 bars
External Design Temperature 0 °C
Maximum Allowable Working Pressure 4 bars
Shop Test Pressure 77.509 bars
Field Test Pressure 64.359 bars
Required Minimum Design Metal Temperature -12.0 °C
Warmest Computed Minimum Design Metal Temperature -48.0 °C
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-105	...	...	K03504	
Nozzle	SA-266 2	...	...	K03506	
Nozzle Flg	SA-105	...	...	K03504	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Leg Baseplate	SA-516 70	...	...	K02700	
Leg Bolting	SA-307 B	...	...		

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
bottom Dish	49.507	0.00	54.000	1.0000	No	No
Shell	49.507	0.00	69.000	1.0000	N/A	No
Shell Flange	49.507	0.00	64.000	1.0000	N/A	No
Dish flange	49.507	0.00	64.000	1.0000	N/A	No
Top dish	49.507	0.00	54.000	1.0000	No	No

Liquid Level: 2253.00 mm. Dens.: 0.000 kg./cm3 Sp. Gr.: 0.001

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Ellipse	50.0	50.0	14.0	11.0	10.1	...	1.00	1.00
Cylinder	2226.0	2176.0	14.0	14.0	10.3	...	1.00	1.00
Body Flg	2341.2	112.0	88.0	86.0	79.3	79.3	1.00	1.00



Body Flg	2453.2	112.0	88.0	86.0	79.3	79.3	1.00	1.00
Ellipse	2503.2	50.0	14.0	11.0	10.1	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
LEGS	WRC-107/537	0.719	n/a	Passed
D1-2"	WRC-107/537	0.135	n/a	Passed
N1-3"	WRC-297	0.288	Shell	Passed
N2-3"	WRC-107/537	0.256	n/a	Passed

Loads for Foundation/Support Design:

Total Wind Shear on top of all Legs	72. Kgf
Total Earthquake Shear on top of all Legs	192. Kgf
Total Wind Moment at top of all Legs	68. Kg-m.
Total Earthquake Moment at top of all Legs	279. Kg-m.
Max. Wind Shear on one Leg (top & bottom)	47. Kgf
Max. Earthq. Shear on one Leg (top & bottom)	127. Kgf
Max. Wind Moment at base of one Leg	85. Kg-m.
Max. Earthquake Moment at base of one Leg	228. Kg-m.
Max. Vertical Load (Wt. + Wind) on one Leg	784. Kgf
Max. Vertical Load (Wt. + Eq.) on one Leg	1468. Kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	1608.0 kg.
Shop Test - Fabricated + water (full)	2250.5 kg.
Shipping - Fab. + rem. intls.+ shipping app.	1608.0 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	1713.0 kg.
Empty - Fab. + intls. + details + wghts.	1713.0 kg.
Operating - Empty + operating liquid (no ca)	1713.5 kg.
Field Test - Empty weight + test medium (full)	2236.3 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
D1-2"	4	...	...	OK	2.50	OK	No Calc[*]
N1-3"		...	...	OK	2.50	OK	No Calc[*]
M1-12"		...	...	...		OK	No Calc[*]
K3-2"		...	...	OK	2.50	OK	No Calc[*]
K1-2"		...	...	OK	2.50	OK	No Calc[*]
K2-2"		...	...	OK	2.50	OK	No Calc[*]
K4-2"		...	...	OK	2.50	OK	No Calc[*]
N2-3"	4	...	...	OK	2.50	OK	No Calc[*]
K11-2"	4	...	...	OK	2.50	OK	No Calc[*]
K12-2"	4	...	...	OK	2.50	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 4 Nozzle : K12-2"
 Minimum MAWP Shells/Flanges : 4 Element : bottom Dish
 Minimum MAPnc Shells/Flanges : 60 Element : Top dish

Computed Vessel M.A.W.P. : 4 bars

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
10	D1-2"	0.000	0.000	125.034
20	N1-3"	1750.000	270.000	142.862
20	M1-12"	800.000	0.000	535.063
20	K3-2"	1000.000	90.000	110.328
20	K1-2"	1000.000	270.000	110.328
20	K2-2"	300.000	270.000	110.328
20	K4-2"	300.000	90.000	110.328
50	N2-3"	0.000	90.000	142.862
50	K11-2"	0.000	225.000	105.600
50	K12-2"	0.000	350.000	105.600

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
D1-2"	2.000 in	160	WNF	2.375	8.738	...	...	213.98	600
K3-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	214.39	600
K1-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	214.39	600
K2-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	214.39	600
K4-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	214.39	600
K11-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	219.51	600
K12-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	219.51	600
N1-3"	3.000 in	160	WNF	3.500	11.125	...	...	267.94	600
N2-3"	3.000 in	160	WNF	3.500	11.125	...	...	273.41	600
M1-12"	12.000 in	160	WNF	12.750	33.325	...	...	272.61	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
D1-2"	SA-105	11.000	10.000	...	...	...
K3-2"	SA-105	12.000	10.000	...	...	...
K1-2"	SA-105	12.000	10.000	...	...	...
K2-2"	SA-105	12.000	10.000	...	...	...
K4-2"	SA-105	12.000	10.000	...	...	...
K11-2"	SA-105	11.000	10.000	...	...	...
K12-2"	SA-105	11.000	10.000	...	...	...
N1-3"	SA-105	11.000	10.000	...	...	...
N2-3"	SA-105	11.000	10.000	...	...	...
M1-12"	SA-266 2	12.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
D1-2"	...	0.0	200.00	0.00	bottom Dish
K3-2"	1000.000	90.0	196.85	0.00	Shell
K1-2"	1000.000	270.0	196.85	0.00	Shell
K2-2"	300.000	270.0	196.85	0.00	Shell
K4-2"	300.000	90.0	196.85	0.00	Shell
K11-2"	...	225.0	196.85	0.00	Top dish
K12-2"	...	350.0	196.85	0.00	Top dish
N1-3"	1750.000	270.0	250.00	0.00	Shell
N2-3"	...	90.0	250.00	0.00	Top dish
M1-12"	800.000	0.0	200.00	0.00	Shell



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 14.0mm. THK X 506.0mm. ID X 50.0mm.	SA-516 70
2	WELD NECK FLANGE: 88.0mm. THK X 718.0mm. OD	SA-266 2
2	CLADDING: 176.500mm X 3.000mm THK	...
1	CLADDING: 2176.000mm X 3.000mm THK	...
3	LEGS, SHAPE: ISA80X80X10 1800.0mm	...
3	BASEPLATES: 154.0mm X 154.0mm X 16.0mm	SA-516 70
3	BASEPLATE BOLTS: 24.0mm	SA-307 B
3	LEG RE-PADS: 150.000mm X 250.000mm X 16.000mm	SA-516 70
2	CLADDING: 114.000mm X 4.500mm THK	...
1	CLASS 600 GR 1.1, 12.0" BLIND FLANGE(S)	SA-105
1	GASKET: 594.0mm. OD X 568.0mm. ID	...
24	BODY FLANGE BOLTS: 36.0mm. DIA	SA-193 B7
48	NUTS FOR BODY FLANGE BOLTS: 36.0mm. DIA	...
1	CYLINDRICAL SEGMENT 14.0mm. THK X 1518.0mm. X 2176.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm. THK X 71.6mm. X 2176.0mm.	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Shell Flange [11]	B	-18	-104	-29	0.053	14.000	1.00	No
Dish flange [11]	B	-18	-104	-29	0.053	14.000	1.00	No
bottom Dish [10]	B	-26	-104	-29	0.150	11.000	1.00	No
bottom Dish [7]	B	-18	-104	-29	0.052	14.000	1.00	No
Shell [20]	B	-18	-48	-29	0.365	14.000	1.00	No
Top dish [10]	B	-26	-104	-29	0.150	11.000	1.00	No
Top dish [7]	B	-18	-104	-29	0.052	14.000	1.00	No
D1-2" [1]	A	-8	-104	-29	0.009	7.645	1.00	No
Nozzle Flg [5]	A	-18	-104					
N1-3" [1]	A	-8	-104	-29	0.013	9.735	1.00	No
Nozzle Flg [5]	A	-18	-104					
M1-12" [1]	B	23	-104		0.015	29.159	1.00	No
Nozzle Flg [5]	A	-18	-104					
K3-2" [1]	A	2	-104		0.052	14.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
K1-2" [1]	A	2	-104		0.052	14.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
K2-2" [1]	A	2	-104		0.052	14.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
K4-2" [1]	A	2	-104		0.052	14.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
N2-3" [1]	A	-8	-104	-29	0.013	9.735	1.00	No
Nozzle Flg [5]	A	-18	-104					
K11-2" [1]	A	-5	-104	-29	0.060	11.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
K12-2" [1]	A	-5	-104	-29	0.060	11.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
Nozl. Bolting [22]		-48			1.000	64.000	1.00	No
Bolting [21]		-48						

Warmest MDMT: 23 -48

Required Minimum Design Metal Temperature -12.0 °C

Warmest Computed Minimum Design Metal Temperature -48.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.



Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (barsUniCode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

KOD for H2 Compressor (175-D)

Design Internal Pressure (for Hydrotest)	49.507	bars
Design Internal Temperature	130.0	°C
Projection of Nozzle from Vessel Top	500	mm.
Projection of Nozzle from Vessel Bottom	500	mm.
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE



Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	10000 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	1.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes



Complete Listing of Vessel Elements and Details:

Element#	1/5
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	bottom Dish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	506 mm.
Element Thickness	11 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	14 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	49.507 bars
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm2
Allowable Stress, Operating	138 N./mm2
Allowable Stress, Hydrotest	234 N./mm2
Material Density	0.00775 kg./cm3
P Number Thickness	32.004 mm.
Yield Stress, Operating	234.4 N./mm2
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	-126.5 mm.
Height/Length of Liquid	176.5 mm.
Liquid Density	0.12979E-05 kg./cm3
Element From Node	10
Detail Type	Lining
Detail ID	Clad
Dist. from "FROM" Node / Offset dist	-126.5 mm.
Height/Length of Lining	176.5 mm.
Thickness of Lining	3 mm.
Density	0.00803 kg./cm3
Element From Node	10
Detail Type	Nozzle
Detail ID	D1-2"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	17.551 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	10
Detail Type	Weight
Detail ID	Vortex breaker
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	10 Kg
Offset from Element Centerline	0 mm.



Element From Node	10
Detail Type	Weight
Detail ID	FP
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	75 Kg
Offset from Element Centerline	0 mm.

Element#	2/5
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	2176 mm.
Inside Diameter	506 mm.
Element Thickness	14 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	14 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	49.507 bars
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1900 mm.
Liquid Density	0.12979E-05 kg./cm3

Element From Node	20
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Lining	2176 mm.
Thickness of Lining	3 mm.
Density	0.00803 kg./cm3

Element From Node	20
Detail Type	Nozzle
Detail ID	N1-3"
Dist. from "FROM" Node / Offset dist	1700 mm.
Nozzle Diameter	3 in.
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	20
Detail Type	Nozzle
Detail ID	M1-12"
Dist. from "FROM" Node / Offset dist	750 mm.
Nozzle Diameter	12 in.
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1



Nozzle Matl	SA-266 2
Element From Node	20
Detail Type	Nozzle
Detail ID	K3-2"
Dist. from "FROM" Node / Offset dist	950 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	20
Detail Type	Nozzle
Detail ID	K1-2"
Dist. from "FROM" Node / Offset dist	950 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	20
Detail Type	Nozzle
Detail ID	K2-2"
Dist. from "FROM" Node / Offset dist	250 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	20
Detail Type	Nozzle
Detail ID	K4-2"
Dist. from "FROM" Node / Offset dist	250 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	20
Detail Type	Leg
Detail ID	LEGS
Dist. from "FROM" Node / Offset dist	250 mm.
Diameter at Leg Centerline	617.82 mm.
Leg Orientation	3
Number of Legs	3
Section Identifier	ISA80X80X10
Length of Legs	1800 mm.
Element From Node	20
Detail Type	Weight
Detail ID	Clip weight
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	30 Kgf



Offset from Element Centerline	0	mm.
Element From Node	20	
Detail Type	Weight	
Detail ID	Demister	
Dist. from "FROM" Node / Offset dist	1600	mm.
Miscellaneous Weight	30	Kgf
Offset from Element Centerline	0	mm.
Element From Node	20	
Detail Type	Weight	
Detail ID	demister suppor	
Dist. from "FROM" Node / Offset dist	1600	mm.
Miscellaneous Weight	30	Kgf
Offset from Element Centerline	0	mm.
Element From Node	20	
Detail Type	Weight	
Detail ID	Nozzles Lining	
Dist. from "FROM" Node / Offset dist	1000	mm.
Miscellaneous Weight	25	Kgf
Offset from Element Centerline	0	mm.
Element From Node	20	
Detail Type	Weight	
Detail ID	Lug wt	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	60	Kgf
Offset from Element Centerline	0	mm.

Element#	3/5
Element From Node	30
Element To Node	40
Element Type	Flange
Description	Shell Flange
Distance "FROM" to "TO"	112 mm.
Flange Inside Diameter	506 mm.
Element Thickness	86 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	88 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	49.507 bars
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm2
Allowable Stress, Operating	138 N./mm2
Allowable Stress, Hydrotest	225 N./mm2
Material Density	0.00775 kg./cm3
P Number Thickness	32.004 mm.
Yield Stress, Operating	222.2 N./mm2
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	30
Detail Type	Lining
Detail ID	Lining
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Lining	114 mm.
Thickness of Lining	4.5 mm.
Density	0.00803 kg./cm3



Element#	4/5
Element From Node	40
Element To Node	50
Element Type	Flange
Description	Dish flange
Distance "FROM" to "TO"	112 mm.
Flange Inside Diameter	506 mm.
Element Thickness	86 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	88 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	49.507 bars
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	40
Detail Type	Lining
Detail ID	Lining
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Lining	114 mm.
Thickness of Lining	4.5 mm.
Density	0.00803 kg./cm3

Element#	5/5
Element From Node	50
Element To Node	60
Element Type	Elliptical
Description	Top dish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	506 mm.
Element Thickness	11 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	14 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	49.507 bars
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm2
Allowable Stress, Operating	138 N./mm2
Allowable Stress, Hydrotest	234 N./mm2
Material Density	0.00775 kg./cm3
P Number Thickness	32.004 mm.
Yield Stress, Operating	234.4 N./mm2
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm.



Height/Length of Liquid	176.5	mm.
Liquid Density	0.11981E-05	kg./cm3

Element From Node	50	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Lining	176.5	mm.
Thickness of Lining	3	mm.
Density	0.00803	kg./cm3

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N2-3"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	K11-2"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	225.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	K12-2"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	350.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
bottom Dish		...	50	...	50
Shell		...	2226	...	2176
Shell Flange		...	2341.2	...	112
Dish flange		...	2453.2	...	112
Top dish		...	2503.2	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

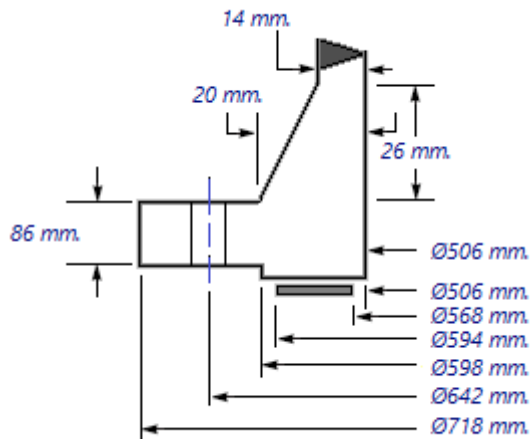
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Flange Input Data Values

Description: FLANGE :

Shell Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	49.51	bars
Design Temperature		130	°C
Internal Corrosion Allowance	ci	1.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	506.000	mm.
Flange Outside Diameter	A	718.000	mm.
Flange Thickness	t	86.0000	mm.
Thickness of Hub at Small End	go	14.0000	mm.
Thickness of Hub at Large End	gl	20.0000	mm.
Length of Hub	h	26.0000	mm.
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm2
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm2
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm2
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm2
Diameter of Bolt Circle	C	642.000	mm.
Nominal Bolt Diameter	a	36.0000	mm.
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	598.000	mm.
Flange Face Inside Diameter	Fid	506.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	594.000	mm.
Gasket Inside Diameter	Gi	568.000	mm.
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm2
Gasket Thickness	tg	3.2000	mm.
Raised face depth	Fd	0.0000	mm.
Axial Force on Flange		250.00	Kgf
Bending Moment on Flange		15.00	Kg-m.





Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (49.51 \cdot (506.0/2 + 1.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 49.51) + Ca$$

$$= 10.3133 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.0) / (254.0 + 0.6 \cdot 13.0)$$

$$= 68.520 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (253.0 + 0.6 \cdot 14.0)$$

$$= 73.904 \text{ bars}$$

Corroded Flange ID,	Bcor = B+2*ci	508.000 mm.
Corroded Large Hub,	g1Cor = g1-ci	19.000 mm.
Corroded Small Hub,	g0Cor = go-ci	13.000 mm.
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	48.000 mm.
Gasket Contact Width,	N = (Go - Gi)/2	13.000 mm.
Basic Gasket Width,	bo = N/2	6.500 mm.
Effective Gasket Width,	b = Cb*sqrt(bo)	6.425 mm.
Gasket Reaction Diameter,	G = Go - 2 * b	581.151 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 581.1509^2 \cdot 49.507$$

$$= 133915.562 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 581.1509 \cdot 3.0 \cdot 49.51$$

$$= 35530.148 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 508.0^2 \cdot 49.507$$

$$= 102324.750 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 133916 - 102325$$

$$= 31590.809 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(133916 + 35530 + 0, 0)$$

$$= 169445.703 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 6.4246 \cdot \pi \cdot 581.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 82468.492 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((169446 + 250 + 4 \cdot 15/581.15) / 172, 82468/172)$$

$$= 96.815 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 36.0 + 6 \cdot 86.0/(3.0 + 0.5)$$

$$= 219.429 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:



$$= C \cdot \sin(\pi / n)$$

$$= 642.0 \cdot \sin(\pi / 24)$$

$$= 83.798 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{83.798 / (2 \cdot 36.0 + 86.0)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	96.815	177.124	
Radial Distance between Hub and Bolts:	47.250	48.000	
Radial Distance between Bolts and Edge:	37.500	38.000	
Circ. Spacing between the Bolts:	82.600	83.798	219.429

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 177.124 \cdot 172.0 / (68.95 \cdot \pi(594.0 + 568.0))$$

$$= 12.103 \text{ mm.}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 177.1236$$

$$= 310647.84 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 169446 - 133916$$

$$= 35530.15 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (642.0 - 581.1509) / 2$$

$$= 30.4245 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((642.0 - 508.0) / 2 + 30.4245)$$

$$= 48.7123 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (642.0 - 508.0 - 19.0) / 2$$

$$= 57.5000 \text{ mm.}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 105.0, Bb = 86.0, Cc = 26.0 \text{ and } Ddg = 16.0 \text{ mm.}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 105.0 \cdot 86.0^3 \cdot (1/3 - 0.210 \cdot (86.0/105.0) \cdot (1 - 1/12 \cdot (86.0/105.0)^4))$$

$$= 1120.557 \text{ cm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 26.0 \cdot 16.0^3 \cdot (1/3 - 0.105 \cdot (16.0/26.0) \cdot (1 - 1/192 \cdot (16.0/26.0)^4))$$

$$= 2.862 \text{ cm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 683.806 + 1.747$$

$$= 1123.419 \text{ cm}^4$$



Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{AR} \right) \left(1 - \frac{1}{12} \left(\frac{t}{AR} \right)^4 \right) \right)$$

$$= 105.0 \cdot 86.0^3 \left(\frac{1}{3} - 0.21 \left(\frac{86.0}{105.0} \right) \left(1 - \frac{1}{12} \left(\frac{86.0}{105.0} \right)^4 \right) \right)$$

$$= 1120.557 \text{ cm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(1123.4, 1120.6)$$

$$= 1123.4 \text{ cm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 3.814 \cdot 13.0^2 \cdot 81.265 \cdot 508.0 / 0.372$$

$$= 625.997 \text{ cm}^4$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading		Force	Distance	Bolt Corr	Moment
End Pressure,	Md	102325.	57.5000	1.0000	5884.
Face Pressure,	Mt	31591.	48.7123	1.0000	1539.
Gasket Load,	Mg	35530.	30.4245	1.0000	1081.
Gasket Seating,	Matm	310648.	30.4245	1.0000	9452.
Moment Load,	Moe	...	...	...	18.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{1}{(0.384 \cdot IP + I)} \right] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$$

$$= 4 \cdot 15 \left[\frac{626.0}{(0.384 \cdot 1123.4 + 626.0)} \right] \cdot (57.5 / (642.0 - 2 \cdot 57.5)) +$$

$$250 \cdot 57.5$$

$$= 18.251 \text{ Kg-m.}$$

Total Moment for Operation, Mo
Total Moment for Gasket seating, Mg

8522. Kg-m.
9452. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 81.265 mm.
Hub Ratio, h/h0 = HL / H0 0.320
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.462

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.879
Factor V per 4.16.5 0.372
Factor f per 4.16.5 1.054

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 718.0 / 508.0$$

$$= 1.413$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$$

$$= (0.66845 + (5.7169 \cdot 1.413^2 \cdot \log_{10}(1.413))) / (1.413^2 - 1) / (1.413 - 1)$$

$$= 5.778$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.413^2(1 + 8.55246 \cdot \log_{10}(1.413)) - 1) / ((1.0472 + 1.9448 \cdot 1.413^2) \cdot (1.413 - 1))$$

$$= 1.748$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.413^2(1 + 8.55246 \cdot \log_{10}(1.413)) - 1) / (1.36136(1.413^2 - 1) \cdot (1.413 - 1))$$

$$= 6.349$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.413^2 + 1) / (1.413^2 - 1)$$

$$= 3.005$$



Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 6.349 \cdot 13.0^2 \cdot 81.265 / 0.372 \\ &= 234705.266 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.879 / 81.265 \\ &= 0.011 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3 / d \\ &= (86.0 \cdot 0.011 + 1) / 1.748 + 86.0^3 / 234705.266 \\ &= 3.814 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (1.0543 \cdot 8522 / 508.0) / (3.8142 \cdot 19.0^2) \\ &= 125.97 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (1.0543 \cdot 9452 / 508.0) / (3.8142 \cdot 19.0^2) \\ &= 139.71 \text{ N./mm}^2 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (2.2408 \cdot 8522 / 508.0) / (3.8142 \cdot 86.0^2) \\ &= 13.07 \text{ N./mm}^2 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (2.2408 \cdot 9452 / 508.0) / (3.8142 \cdot 86.0^2) \\ &= 14.49 \text{ N./mm}^2 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (5.7779 \cdot 8522 / (86.0^2 \cdot 508.0)) - 3.0047 \cdot 13 \\ &= 89.26 \text{ N./mm}^2 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned} &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\ &= (5.7779 \cdot 9452 / (86.0^2 \cdot 508.0)) - 3.0047 \cdot 14 \\ &= 98.99 \text{ N./mm}^2 \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned} &= (SHo + \max(SRo, STo)) / 2 \\ &= (126 + \max(13, 89)) / 2 \\ &= 107.61 \text{ N./mm}^2 \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned} &= (SHa + \max(SRa, STa)) / 2 \\ &= (140 + \max(14, 99)) / 2 \\ &= 119.35 \text{ N./mm}^2 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned} &= (Wm_1 + Fa + 4Me/G) / Ab \\ &= 169799 / 177.1236 \\ &= 94.01 \text{ N./mm}^2 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (Wm_2 / Ab) \\ &= (82468 / 177.1236) \\ &= 45.66 \text{ N./mm}^2 \end{aligned}$$

Flange Stress Analysis Results: N./mm²



	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	125.97	207.00	139.71	207.00
Radial Flange	13.07	138.00	14.49	138.00
Tangential Flange	89.26	138.00	98.99	138.00
Maximum Average	107.61	138.00	119.35	138.00
Bolting	94.01	172.00	45.66	172.00

Minimum Required Flange Thickness 79.350 mm.

Estimated M.A.W.P. (Operating) 64 bars

Estimated Finished Weight of Flange at given Thk. 141.5 kg.

Estimated Unfinished Weight of Forging at given Thk 176.9 kg.

Flange Stress results per WRC-538 at computed Sfmax (N/mm2):

		Actual	Operating Allowed	stress ratio	Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	449.56	499.99	0.899	449.56	499.99	0.899
Tangential Flange	St	320.70	374.99	0.855	320.70	374.99	0.855
Tangential Flange	Sto	137.16	250.00	0.549	137.16	250.00	0.549
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	367.65	749.99	0.490	367.65	749.99	0.490
Combined	Sh + Sto	586.72	749.99	0.782	586.72	749.99	0.782
Radial Flange	Sr	46.95	499.99	0.094	46.95	499.99	0.094
Maximum Stress Ratio				1.000			1.000



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1num / F1den) \\
= \max(1; 1.3479/1.2871) \\
= 1.0472$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3) \\
= \max(0.25; 0.3982/1.2464) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 0.2543/0.7671) \\
= 0.3195$$

In this case, the hub ratio $h/h0$, 0.3199 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sb_{max}	507.0 N./mm ²
Minimum permissible bolt stress	Sb_{min}	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sf_{max}	557.2 N./mm ²
Target assembly gasket stress	SgT	172.0 N./mm ²
Maximum permissible gasket stress	Sg_{max}	380.0 N./mm ²
Minimum gasket operating stress	Sg_{min-0}	97.0 N./mm ²
Minimum gasket seating stress	Sg_{min-S}	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	$Phig$	0.700
Root Area of a Single Bolt	Abs	7.3802 cm ²
Max. permissible single flange rotation for gasket	$Thetamax$	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	222.2 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.456 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (594.0^2 - 568.0^2)$$

$$= 237.3 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172.0 \cdot 237.285 / (24 \cdot 7.38)$$

$$= 230.4 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(230.421; 507.0)$$

$$= 230.4 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(230.421; 290.0)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290.0; 557.211)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 290.0 \geq 140.0(237.285 / (24 \cdot 7.38))$$

$$:: 290.0 \geq 187.6 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 290.0 \geq (97.0 \cdot 237.285 + \pi/4 \cdot 49.507 \cdot 568.0^2) / (0.7 \cdot 7.38 \cdot 24)$$

$$:: 290.0 \geq 286.8 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 290.0 \leq 380.0(237.285 / (24 \cdot 7.38))$$

$$:: 290.0 \leq 509.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\text{thetamax} / \text{thetafmax})$$

$$:: 290.0 \leq 557.211(1.0/0.456)$$

$$:: 290.0 \leq 1221.8 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 290.0 \cdot 0.2 \cdot 7.38 \cdot 36.0$$

$$= 157.1 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 107 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 9451.5 / 1.0 \cdot 9806.637 \cdot 0.372 / (3.19 \cdot 202713 \cdot 13.0^2 \cdot 81.265 \cdot 0.3)$$

$$= 0.673 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$



$$= 52.14 \cdot 8521.9 / 1.0 \cdot 9806.637 \cdot 0.372 / (3.19 \cdot 196301 \cdot 13.0^2 \cdot 81.265 \cdot 0.3)$$

$$= 0.627 \quad (\text{should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 9451.5 / 1.0 \cdot 9806.637 \cdot 0.372 / (3.814 \cdot 202713 \cdot 13.0^2 \cdot 81.265 \cdot 0.3)$$

$$= 0.563 \quad (\text{should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 8521.9 / 1.0 \cdot 9806.637 \cdot 0.372 / (3.814 \cdot 196301 \cdot 13.0^2 \cdot 81.265 \cdot 0.3)$$

$$= 0.524 \quad (\text{should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.053, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	125.55	207.00	138.27	207.00
Radial Flange	14.41	138.00	15.87	138.00
Tangential Flange	85.67	138.00	94.35	138.00
Maximum Average	105.61	138.00	116.31	138.00
Bolting	94.01	172.00	45.66	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 77.876 mm.

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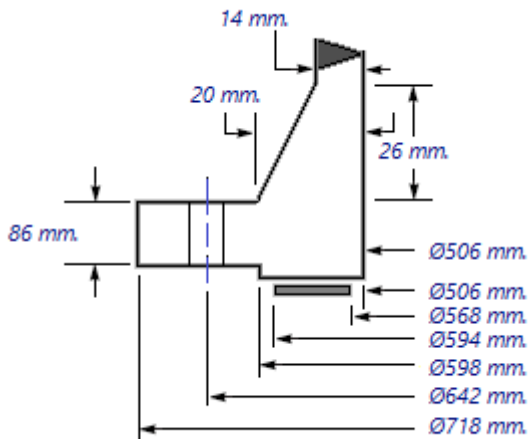


Flange Input Data Values

Description: New Flange :

Dish flange

Flange Type	Integral Weld Neck		
Design Pressure	P	49.51	bars
Design Temperature		130	°C
Internal Corrosion Allowance	ci	1.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	506.000	mm.
Flange Outside Diameter	A	718.000	mm.
Flange Thickness	t	86.0000	mm.
Thickness of Hub at Small End	go	14.0000	mm.
Thickness of Hub at Large End	gl	20.0000	mm.
Length of Hub	h	26.0000	mm.
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm2
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm2
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm2
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm2
Diameter of Bolt Circle	C	642.000	mm.
Nominal Bolt Diameter	a	36.0000	mm.
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	598.000	mm.
Flange Face Inside Diameter	Fid	506.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	594.000	mm.
Gasket Inside Diameter	Gi	568.000	mm.
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm2
Gasket Thickness	tg	3.1750	mm.
Raised face depth	Fd	0.0000	mm.
Axial Force on Flange		75.00	Kgf
Bending Moment on Flange		5.00	Kg-m.





Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (49.51 \cdot (506.0/2 + 1.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 49.51) + Ca$$

$$= 10.3133 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.0) / (254.0 + 0.6 \cdot 13.0)$$

$$= 68.520 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (253.0 + 0.6 \cdot 14.0)$$

$$= 73.904 \text{ bars}$$

Corroded Flange ID,	Bcor = B+2*ci	508.000 mm.
Corroded Large Hub,	g1Cor = g1-ci	19.000 mm.
Corroded Small Hub,	g0Cor = go-ci	13.000 mm.
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	48.000 mm.
Gasket Contact Width,	N = (Go - Gi)/2	13.000 mm.
Basic Gasket Width,	bo = N/2	6.500 mm.
Effective Gasket Width,	b = Cb*sqrt(bo)	6.425 mm.
Gasket Reaction Diameter,	G = Go - 2 * b	581.151 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 581.1509^2 \cdot 49.507$$

$$= 133915.562 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 581.1509 \cdot 3.0 \cdot 49.51$$

$$= 35530.148 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 508.0^2 \cdot 49.507$$

$$= 102324.750 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 133916 - 102325$$

$$= 31590.809 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(133916 + 35530 + 0, 0)$$

$$= 169445.703 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 6.4246 \cdot \pi \cdot 581.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 82468.492 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((169446 + 75 + 4 \cdot 5 / 581.15) / 172, 82468 / 172)$$

$$= 96.676 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 36.0 + 6 \cdot 86.0 / (3.0 + 0.5)$$

$$= 219.429 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:



$$= C \cdot \sin(\pi / n)$$

$$= 642.0 \cdot \sin(\pi / 24)$$

$$= 83.798 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{83.798}{(2 \cdot 36.0 + 86.0)}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	96.676	177.124	
Radial Distance between Hub and Bolts:	47.250	48.000	
Radial Distance between Bolts and Edge:	37.500	38.000	
Circ. Spacing between the Bolts:	82.600	83.798	219.429

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 177.124 \cdot 172.0 / (68.95 \cdot \pi(594.0 + 568.0))$$

$$= 12.103 \text{ mm.}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 177.1236$$

$$= 310647.84 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 169446 - 133916$$

$$= 35530.15 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (642.0 - 581.1509) / 2$$

$$= 30.4245 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((642.0 - 508.0) / 2 + 30.4245)$$

$$= 48.7123 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (642.0 - 508.0 - 19.0) / 2$$

$$= 57.5000 \text{ mm.}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 105.0, Bb = 86.0, Cc = 26.0 \text{ and } Ddg = 16.0 \text{ mm.}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 105.0 \cdot 86.0^3 \cdot (1/3 - 0.210 \cdot (86.0/105.0) \cdot (1 - 1/12 \cdot (86.0/105.0)^4))$$

$$= 1120.557 \text{ cm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 26.0 \cdot 16.0^3 \cdot (1/3 - 0.105 \cdot (16.0/26.0) \cdot (1 - 1/192 \cdot (16.0/26.0)^4))$$

$$= 2.862 \text{ cm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 683.806 + 1.747$$

$$= 1123.419 \text{ cm}^4$$



Flange Inertia Term [lp,term2]:
 $= Ar \cdot t^3 (1/3 - 0.21 (t/AR) (1 - 1/12 (t/AR)^4))$
 $= 105.0 \cdot 86.0^3 (1/3 - 0.21 (86.0/105.0) (1 - 1/12 (86.0/4.134)^4))$
 $= 1120.557 \text{ cm}^4$

Flange Inertia [lp]:
 $= \max (lp,term1, lp,term2)$
 $= \max (1123.4, 1120.6)$
 $= 1123.4 \text{ cm}^4$

Flange Moment of Inertia for Integral Types [I]:
 $= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$
 $= 0.0874 \cdot 3.814 \cdot 13.0^2 \cdot 81.265 \cdot 508.0 / 0.372$
 $= 625.997 \text{ cm}^4$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading		Force	Distance	Bolt Corr	Moment
End Pressure,	Md	102325.	57.5000	1.0000	5884.
Face Pressure,	Mt	31591.	48.7123	1.0000	1539.
Gasket Load,	Mg	35530.	30.4245	1.0000	1081.
Gasket Seating,	Matm	310648.	30.4245	1.0000	9452.
Moment Load,	Moe	...	...	...	6.

Moment Value, eq. 4.16.16, [Moe]:
 $= 4Me [1 / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$
 $= 4 \cdot 5 [626.0 / (0.384 \cdot 1123.4 + 626.0)] \cdot (57.5 / (642.0 - 2 \cdot 57.5)) +$
 $75 \cdot 57.5$
 $= 5.604 \text{ Kg-m.}$

Total Moment for Operation, Mo 8509. Kg-m.
 Total Moment for Gasket seating, Mg 9452. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 81.265 mm.
 Hub Ratio, h/h0 = HL / H0 0.320
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.462

Flange Factors for Integral Flange:
 Factor F per 4.16.5 0.879
 Factor V per 4.16.5 0.372
 Factor f per 4.16.5 1.054

Stress factor from Table 4.16.4 [K]:
 $= A / B$
 $= 718.0 / 508.0$
 $= 1.413$

Stress factor from Table 4.16.4 [Y]:
 $= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169 (1.413^2 \cdot \log_{10}(1.413))) / (1.413^2 - 1)) / (1.413 - 1)$
 $= 5.778$

Stress factor from Table 4.16.4 [T]:
 $= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.413^2 (1 + 8.55246 \cdot \log_{10}(1.413)) - 1) / ((1.0472 + 1.9448 \cdot 1.413^2) \cdot (1.413 - 1))$
 $= 1.748$

Stress factor from Table 4.16.4 [U]:
 $= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.413^2 (1 + 8.55246 \cdot \log_{10}(1.413)) - 1) / (1.36136 (1.413^2 - 1) \cdot (1.413 - 1))$
 $= 6.349$

Stress factor from Table 4.16.4 [Z]:
 $= (K^2 + 1) / (K^2 - 1)$
 $= (1.413^2 + 1) / (1.413^2 - 1)$
 $= 3.005$



Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned}
 &= U \cdot g_0^2 \cdot h_0 / V \\
 &= 6.349 \cdot 13.0^2 \cdot 81.265 / 0.372 \\
 &= 234705.266 \text{ mm}^3
 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned}
 &= F / h_0 \\
 &= 0.879 / 81.265 \\
 &= 0.011 \text{ 1/mm}
 \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned}
 &= (t \cdot e + 1) / T + t^3 / d \\
 &= (86.0 \cdot 0.011 + 1) / 1.748 + 86.0^3 / 234705.266 \\
 &= 3.814
 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned}
 &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\
 &= (1.0543 \cdot 8509 / 508.0) / (3.8142 \cdot 19.0^2) \\
 &= 125.78 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned}
 &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\
 &= (1.0543 \cdot 9452 / 508.0) / (3.8142 \cdot 19.0^2) \\
 &= 139.71 \text{ N./mm}^2
 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned}
 &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\
 &= (2.2408 \cdot 8509 / 508.0) / (3.8142 \cdot 86.0^2) \\
 &= 13.05 \text{ N./mm}^2
 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned}
 &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\
 &= (2.2408 \cdot 9452 / 508.0) / (3.8142 \cdot 86.0^2) \\
 &= 14.49 \text{ N./mm}^2
 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned}
 &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\
 &= (5.7779 \cdot 8509 / (86.0^2 \cdot 508.0)) - 3.0047 \cdot 13 \\
 &= 89.12 \text{ N./mm}^2
 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned}
 &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\
 &= (5.7779 \cdot 9452 / (86.0^2 \cdot 508.0)) - 3.0047 \cdot 14 \\
 &= 98.99 \text{ N./mm}^2
 \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned}
 &= (SHo + \max(SRo, STo)) / 2 \\
 &= (126 + \max(13, 89)) / 2 \\
 &= 107.45 \text{ N./mm}^2
 \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned}
 &= (SHa + \max(SRa, STa)) / 2 \\
 &= (140 + \max(14, 99)) / 2 \\
 &= 119.35 \text{ N./mm}^2
 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned}
 &= (Wm1 + Fa + 4Me/G) / Ab \\
 &= 169555 / 177.1236 \\
 &= 93.88 \text{ N./mm}^2
 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned}
 &= (Wm2 / Ab) \\
 &= (82468 / 177.1236) \\
 &= 45.66 \text{ N./mm}^2
 \end{aligned}$$

Flange Stress Analysis Results: N./mm²



	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	125.78	207.00	139.71	207.00
Radial Flange	13.05	138.00	14.49	138.00
Tangential Flange	89.12	138.00	98.99	138.00
Maximum Average	107.45	138.00	119.35	138.00
Bolting	93.88	172.00	45.66	172.00

Minimum Required Flange Thickness 79.350 mm.

Estimated M.A.W.P. (Operating) 64 bars

Estimated Finished Weight of Flange at given Thk. 141.5 kg.

Estimated Unfinished Weight of Forging at given Thk 176.9 kg.

Flange Stress results per WRC-538 at computed Sfmax (N/mm2):

		Actual	Operating Allowed	stress ratio	Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	449.56	499.99	0.899	449.56	499.99	0.899
Tangential Flange	St	320.70	374.99	0.855	320.70	374.99	0.855
Tangential Flange	Sto	137.16	250.00	0.549	137.16	250.00	0.549
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	367.65	749.99	0.490	367.65	749.99	0.490
Combined	Sh + Sto	586.72	749.99	0.782	586.72	749.99	0.782
Radial Flange	Sr	46.95	499.99	0.094	46.95	499.99	0.094
Maximum Stress Ratio				1.000			1.000

Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1num / F1den) \\
= \max(1; 1.3479/1.2871) \\
= 1.0472$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3) \\
= \max(0.25; 0.3982/1.2464) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 0.2543/0.7671) \\
= 0.3195$$

In this case, the hub ratio $h/h0$, 0.3199 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.0 N./mm2
Minimum permissible bolt stress	Sbmin	290.0 N./mm2
Max. permissible bolt stress prior to flange damage	Sfmax	557.2 N./mm2
Target assembly gasket stress	SgT	172.0 N./mm2
Maximum permissible gasket stress	Sgmax	380.0 N./mm2
Minimum gasket operating stress	Sgmin-0	97.0 N./mm2
Minimum gasket seating stress	Sgmin-S	140.0 N./mm2
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	7.3802 cm2
Max. permissible single flange rotation for gasket	Thetamax	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	222.2 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.456 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (594.0^2 - 568.0^2)$$

$$= 237.3 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172.0 \cdot 237.285 / (24 \cdot 7.38)$$

$$= 230.4 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(230.421; 507.0)$$

$$= 230.4 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(230.421; 290.0)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290.0; 557.211)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 290.0 \geq 140.0(237.285 / (24 \cdot 7.38))$$

$$:: 290.0 \geq 187.6 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 290.0 \geq (97.0 \cdot 237.285 + \pi/4 \cdot 49.507 \cdot 568.0^2) / (0.7 \cdot 7.38 \cdot 24)$$

$$:: 290.0 \geq 286.8 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 290.0 \leq 380.0(237.285 / (24 \cdot 7.38))$$

$$:: 290.0 \leq 509.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(thetagmax / thetafmax)$$

$$:: 290.0 \leq 557.211(1.0 / 0.456)$$

$$:: 290.0 \leq 1221.8 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 290.0 \cdot 0.2 \cdot 7.38 \cdot 36.0$$

$$= 157.1 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 107 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 9451.5 / 1.0 \cdot 9806.637 \cdot 0.372 / (3.19 \cdot 202713 \cdot 13.0^2 \cdot 81.265 \cdot 0.3)$$

$$= 0.673 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$



$$= 52.14 \cdot 8509.3 / 1.0 \cdot 9806.637 \cdot 0.372 / (3.19 \cdot 196301 \cdot 13.0^2 \cdot 81.265 \cdot 0.3)$$

$$= 0.626 \quad (\text{should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 9451.5 / 1.0 \cdot 9806.637 \cdot 0.372 / (3.814 \cdot 202713 \cdot 13.0^2 \cdot 81.265 \cdot 0.3)$$

$$= 0.563 \quad (\text{should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 8509.3 / 1.0 \cdot 9806.637 \cdot 0.372 / (3.814 \cdot 196301 \cdot 13.0^2 \cdot 81.265 \cdot 0.3)$$

$$= 0.524 \quad (\text{should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.053, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	125.36	207.00	138.27	207.00
Radial Flange	14.39	138.00	15.87	138.00
Tangential Flange	85.55	138.00	94.35	138.00
Maximum Average	105.45	138.00	116.31	138.00
Bolting	93.88	172.00	45.66	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 77.876 mm.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
bottom Dish		49.507	14	1	506	138	1
Shell		49.507	14	1	506	138	2
Shell Flange		49.507	88	1	506	138	3
Dish flange		49.507	88	1	506	138	4
Top dish		49.507	14	1	506	138	5

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.W. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
bottom Dish		49.507	54.396	59.7357	11	10.098	1
Shell		49.507	68.5201	73.9041	14	10.3134	2
Shell Flange		49.507	63.5179	64.7215	86	79.3496	3
Dish flange		49.507	63.5902	64.7953	86	79.3496	4
Top dish		49.507	54.3962	59.7357	11	10.0979	5

Minimum 4 60

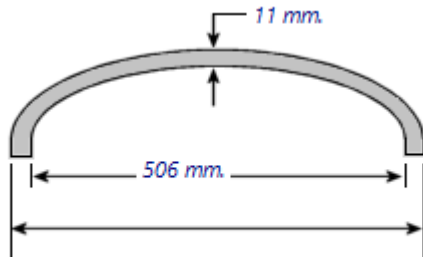
MAWP: 4 bars, limited by: DESIGN (user specified)

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: bottom Dish

Elliptical Head Design Information:					
Design Pressure	P	49.507 bars	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope) S		138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb) Sa		138.0 N./mm^2
Inside Diameter	Di	506.000 mm.	Actual Stress		125.6 N./mm^2
Spec. Min. Thk t,e, or sa		11.0000 mm.	Spec. Nominal Thk.		14.000 mm.
Surface Area		3929.9 cm^2	SG of Contents		0.001
Element Volume		27018.0 cm3	Weight of Contents		0.0 Kg
Empty Weight		44.0 Kg	Operating Weight		44.0 Kg
			Comp. All. Ope. Stress		142.5 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.325 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.163 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (49.507 \cdot 508.0 \cdot 0.995) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 49.507)$$

$$= 9.0980 + 1.0000 = 10.0980 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 10.0) / (0.995 \cdot 508.0 + 0.2 \cdot 10.0)$$

$$= 54 - 0 = 54 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 11.0) / (1.0 \cdot 506.0 + 0.2 \cdot 11.0)$$

$$= 60 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (49.507 \cdot (0.995 \cdot 508.0 + 0.2 \cdot 10.0)) / (2 \cdot 1.0 \cdot 10.0)$$

$$= 125.595 \text{ N./mm}^2$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (49.507 \cdot 254.0) / (138.0 \cdot 1.0 - 0.6 \cdot 49.507) + 1.0$$

$$= 10.313 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.000 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.0) / (254.0 + 0.6 \cdot 13.0)$$

$$= 69 - 0 = 69 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (508.0 / (2 \cdot 127.5)))^2 / 6$$

$$= 0.994782$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 11.288 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 11.0, tr = 1.5, c = 1.0 mm., E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.15$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 14.0, tr = 0.675, c = 1.0 mm., E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.052$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.660 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	11.000 mm.	Basic MDMT	-26 °C
Stress/Thk. Ratio	0.150	Temperature Difference	78 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

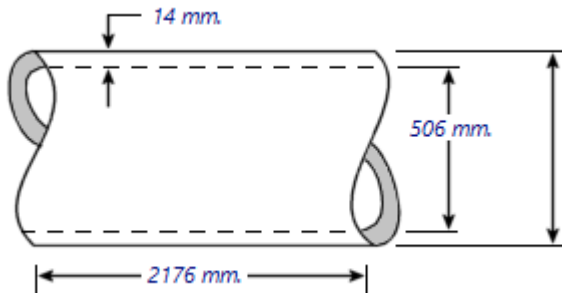
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.325 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.163 mm.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell

Cylindrical Shell Design Information:					
Design Pressure	P	49.507 bars	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2176.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	506.000 mm.	Actual Stress		99.7 N./mm^2
Spec. Min. Thk t,e, or sa		14.0000 mm.	Spec. Nominal Thk.		14.000 mm.
Surface Area		36504.8 cm^2	SG of Contents		0.001
Element Volume		437650.8 cm3	Weight of Contents		0.5 Kg
Empty Weight		385.7 Kg	Operating Weight		386.2 Kg
			Comp. All. Ope. Stress		145.7 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (49.507 \cdot 254.0) / (138.0 \cdot 1.0 \cdot 0.6 \cdot 49.507)$$

$$= 9.3134 + 1.0000 = 10.3134 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.0) / (254.0 + 0.6 \cdot 13.0)$$

$$= 69 - 0 = 69 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (253.0 + 0.6 \cdot 14.0)$$

$$= 74 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (49.507 \cdot (254.0 + 0.6 \cdot 13.0)) / (1.0 \cdot 13.0)$$

$$= 99.706 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 2.692 %

Minimum Design Metal Temperature Results:



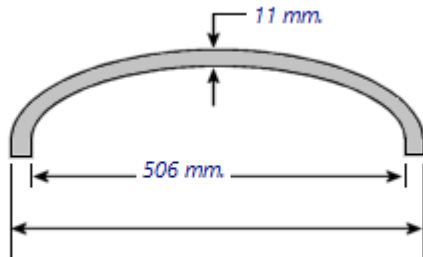
Govern. thk, tg = 14.0, tr = 4.745, c = 1.0 mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.365$, Temp. Reduction = 67 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.660 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm.	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.365	Temperature Difference	30 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Top dish

Elliptical Head Design Information:					
Design Pressure	P	49.507 bars	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope) S		138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb) Sa		138.0 N./mm^2
Inside Diameter	Di	506.000 mm.	Actual Stress		125.6 N./mm^2
Spec. Min. Thk t _e , or sa		11.0000 mm.	Spec. Nominal Thk.		14.000 mm.
Surface Area		3929.9 cm^2	SG of Contents		0.001
Element Volume		27018.0 cm3	Weight of Contents		0.0 Kg
Empty Weight		44.0 Kg	Operating Weight		44.0 Kg
			Comp. All. Ope. Stress		142.5 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.325 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.163 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (49.507 \cdot 508.0 \cdot 0.995) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 49.507)$$

$$= 9.0979 + 1.0000 = 10.0979 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 10.0) / (0.995 \cdot 508.0 + 0.2 \cdot 10.0)$$

$$= 54 - 0.14879E-04 = 54 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 11.0) / (1.0 \cdot 506.0 + 0.2 \cdot 11.0)$$

$$= 60 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (49.507 \cdot (0.995 \cdot 508.0 + 0.2 \cdot 10.0)) / (2 \cdot 1.0 \cdot 10.0)$$

$$= 125.594 \text{ N./mm}^2$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (49.507 \cdot 254.0) / (138.0 \cdot 1.0 - 0.6 \cdot 49.507) + 1.0$$

$$= 10.313 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 0.000 bars](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.0) / (254.0 + 0.6 \cdot 13.0)$$

$$= 69 - 0.14879E-04 = 69 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (508.0 / (2 \cdot 127.5)))^2 / 6$$

$$= 0.994782$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 11.288 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 11.0, tr = 1.5, c = 1.0 mm., E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.15$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 14.0, tr = 0.675, c = 1.0 mm., E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.052$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.660 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	11.000 mm.	Basic MDMT	-26 °C
Stress/Thk. Ratio	0.150	Temperature Difference	78 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.325 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.163 mm.

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Hydrostatic Test Pressure Results:

UG99(b) = 1.30 * M.A.W.P. * Sa/S 4.758 bars
 UG99b(b)(35) = 1.30 * Design Pres * Sa/S 64.359 bars
 UG99(c) = 1.30 * M.A.P. - Head(Hyd) 77.509 bars
 UG100 = 1.10 * M.A.W.P. * Sa/S 4.026 bars
 PED = max(1.43*DP, 1.25*DP*ratio) 70.795 bars
 PED&UG99(b)(35) = max(1.43*DP, 1.3*DP*ratio) 70.795 bars
 App 27-4 = M.A.W.P. 3.660 bars

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
bottom Dish	179.2	234.0	0.766	77.56
Shell	144.8	234.0	0.619	77.56
Top dish	179.2	234.0	0.766	77.56

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D1-2"	Nozzle	138.00	138.00	1.000
N1-3"	Nozzle	138.00	138.00	1.000
M1-12"	Nozzle	138.00	138.00	1.000
K3-2"	Nozzle	138.00	138.00	1.000
K1-2"	Nozzle	138.00	138.00	1.000
K2-2"	Nozzle	138.00	138.00	1.000
K4-2"	Nozzle	138.00	138.00	1.000
N2-3"	Nozzle	138.00	138.00	1.000
K11-2"	Nozzle	138.00	138.00	1.000
K12-2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
bottom Dish	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Shell Flange	138.00	138.00	1.000
Dish flange	138.00	138.00	1.000
Top dish	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
D1-2"	27.50	225.00	0.122
N1-3"	32.31	225.00	0.144
M1-12"	39.97	225.00	0.178
K3-2"	16.41	225.00	0.073
K1-2"	16.41	225.00	0.073
K2-2"	16.41	225.00	0.073
K4-2"	16.41	225.00	0.073
N2-3"	32.31	225.00	0.144
K11-2"	16.41	225.00	0.073
K12-2"	16.41	225.00	0.073



Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	4.758 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	64.359 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	77.509 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	4.026 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	70.795 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	70.795 bars
App 27-4	= M.A.W.P.	3.660 bars

UG-99(b) Note 35, Test Pressure Calculation:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 49.507 • 1.0
 = 64.359 bars

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm2 & bars):

From To	Stress	Allowable	Ratio	Pressure
bottom Dish	163.5	234.0	0.699	64.46
Shell	129.8	234.0	0.555	64.46
Top dish	163.5	234.0	0.699	64.46

Stress ratios for Nozzle and Pad Materials (N./mm2):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D1-2"	Nozzle	138.00	138.00	1.000
N1-3"	Nozzle	138.00	138.00	1.000
M1-12"	Nozzle	138.00	138.00	1.000
K3-2"	Nozzle	138.00	138.00	1.000
K1-2"	Nozzle	138.00	138.00	1.000
K2-2"	Nozzle	138.00	138.00	1.000
K4-2"	Nozzle	138.00	138.00	1.000
N2-3"	Nozzle	138.00	138.00	1.000
K11-2"	Nozzle	138.00	138.00	1.000
K12-2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm2):

Description	Ambient	Operating	Ratio
bottom Dish	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Shell Flange	138.00	138.00	1.000
Dish flange	138.00	138.00	1.000
Top dish	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm2):

Description	Ambient	Operating	Ratio
D1-2"	22.85	225.00	0.102
N1-3"	26.86	225.00	0.119
M1-12"	33.22	225.00	0.148
K3-2"	13.63	225.00	0.061
K1-2"	13.63	225.00	0.061



K2-2"	13.63	225.00	0.061
K4-2"	13.63	225.00	0.061
N2-3"	26.86	225.00	0.119
K11-2"	13.63	225.00	0.061
K12-2"	13.63	225.00	0.061

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	528	10	0.0026305	109.478
20	30	2268.17	534	13	0.0010564	86.9699
30	40	No Calc	...	85	No Calc	No Calc
40	50	No Calc	...	85	No Calc	No Calc
50	60	No Calc	528	10	0.0026305	109.478

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	11	No Calc	...	23.0369
20	30	14	No Calc	...	28.2283
30	40	86	79.3496	...	No Calc
40	50	86	79.3496	...	No Calc
50	60	11	No Calc	...	23.0369

Minimum

23

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	2268.17	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	43.979	27018	40.8377	27299.5	3.07853
20	30	385.718	437651	358.855	441117	27.0002
30	40	144.63	23068.6	143.223	23110	10.1241
40	50	144.63	23068.6	143.223	23110	10.1241
50	60	43.979	27018	40.8377	27299.5	3.07853
Total		762	537823.81	726	541936.38	53

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Liqd	0.034019	...	-63.25	...	Liquid: 10
10	Lini	9.20895	...	-38.25	...	Clad
10	Nozl	18.7796	...	-613.039	...	D1-2"
10	Wght	10	...	...	...	Vortex breaker
10	Wght	75	...	...	...	FP
20	Liqd	0.48477	...	950	...	Liquid: 20
20	Lini	82.9319	...	1088	...	clad
20	Nozl	19.5315	-378.006	1700	...	N1-3"
20	Nozl	318.467	...	750	-353	M1-12"
20	Nozl	11.5754	351.425	950	...	K3-2"
20	Nozl	11.5754	-351.425	950	...	K1-2"
20	Nozl	11.5754	-351.425	250	...	K2-2"
20	Nozl	11.5754	351.425	250	...	K4-2"
20	Legs	76.9356	...	-650	...	LEGS
20	Wght	30	...	...	...	Clip weight
20	Wght	30	...	1600	...	Demister
20	Wght	30	...	1600	...	demister suppor
20	Wght	25	...	1000	...	Nozzles Lining
20	Wght	60	...	...	...	Lug wt
30	Lini	6.49773	...	57	...	Lining
40	Lini	6.49773	...	57	...	Lining
50	Liqd	0.031398	...	113.25	...	Liquid: 30
50	Lini	9.20895	...	88.25	...	Clad
50	Nozl	19.2983	100	285.705	...	N2-3"
50	Nozl	11.4869	-70.7107	259.124	70.7107	K11-2"
50	Nozl	11.4869	-17.3648	259.124	-98.4808	K12-2"

Total Weight of Each Detail Type:

Liquid	0.6
Cladding & Overlay	114.3
Nozzles	445.4
Legs	76.9
Weights	260.0

Sum of the Detail Weights 897.2 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	816.3	816.3	816.3	816.3	816.3	816.3
Cladding	114.3	114.3	114.3	114.3	114.3	114.3
Nozzles	445.4	445.4	445.4	445.4	445.4	445.4
Legs	76.9	76.9	76.9	76.9	76.9	76.9
Wld Weights	155.0	155.0	155.0	155.0	155.0	155.0
Empty Weights	...	...	...	105.0	105.0	...



Test Weights	...	105.0	...	...	...	...
Ope Weights	...	...	...	...	...	105.0
Ope. Liquid	...	...	...	...	...	0.6
Test Liquid	...	537.5	...	...	...	...
Totals	1608.0	2250.5	1608.0	1713.0	1713.0	1713.5

Field Installation Options:

* Lining installed after lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	1608.0 kg.
Shop Test Wt.	- Fabricated weight + water (full)	2250.5 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	1608.0 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	1713.0 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	1713.0 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	1713.5 kg.
Field Test Wt.	- Empty weight + test medium (full)	2236.3 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	553.9 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	647.8 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	567.5 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	3929.85	3570.22
20	30	36504.8	34590.7
30	40	4468.98	1812.2
40	50	4468.98	1812.2
50	60	3929.85	3570.22

Total	53302.5	45355.5 cm ²
Total	57.4	48.8 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	150.837	160.08	187.048	...	160.046
20	Legs	111.675	121.259	171.454	112.421	121.203
Legs	30	860.343	934.176	1320.88	...	933.747
30	40	154.754	161.252	184.307	...	161.252
40	50	154.754	161.252	184.307	...	161.252
50	60	89.3297	98.5701	125.54	0.319	98.5387

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	...	...	...
20	Legs	-160.046	-160.08	-187.048



Legs	30	1354.79	1355.25	1815.03
30	40	421.043	421.074	494.153
40	50	259.791	259.822	309.847
50	60	98.5387	98.5701	125.54

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	...	...	...
20	Legs	112.421	112.421	112.421
Legs	30	0.319	0.319	0.319
30	40	0.319	0.319	0.319
40	50	0.319	0.319	0.319
50	60	0.319	0.319	0.319

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
D1-2"	91.42	102.00	130	600	GR 1.1	...	...	...	...
N1-3"	91.42	102.00	130	600	GR 1.1	53.539	83.592	91.421	83.592
M1-12"	91.42	102.00	130	600	GR 1.1	...	...	...	...
K3-2"	91.42	102.00	130	600	GR 1.1	...	...	...	...
K1-2"	91.42	102.00	130	600	GR 1.1	...	...	...	...
K2-2"	91.42	102.00	130	600	GR 1.1	...	...	...	...
K4-2"	91.42	102.00	130	600	GR 1.1	...	...	...	...
N2-3"	91.42	102.00	130	600	GR 1.1	...	...	...	...
K11-2"	91.42	102.00	130	600	GR 1.1	...	...	...	...
K12-2"	91.42	102.00	130	600	GR 1.1	...	...	...	...
Min Rating	83.592	102.000, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
N1-3"	111.125	0.50	130	612	3.660	83.592	Pass

Evaluating Flange on Nozzle [N1-3"]:
 $= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$
 $= 16(130) + 4 \cdot \text{abs}(612)111.125 \leq \pi \cdot 111.125^3 [(91.421 - 3.660) + .500 \cdot 91.421]$
 $= 0.23538E+04 \leq 0.58678E+04$ Check Passes

Maximum Pressure for Flange on Nozzle N1-3" [Pmax]:
 $= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(91.421, 0.500 \cdot 91.421 + 91.421 - 2353.8 / (\pi \cdot 111.125^3))$
 $= \min(91.421, 83.592)$
 $= 83.592 \text{ bars}$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 8.27919 Hz.

The Natural Frequency for the Vessel (Ope...) is 8.27802 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	10000 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.228 \cdot 0.972)/(1 + 1.7 \cdot 3.4 \cdot 0.228)))$
 $= \min(0.85, 0.91)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	8.278 Hz
Natural Frequency of Vessel (Empty)	8.279 Hz
Natural Frequency of Vessel (Test)	7.257 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.567
Structure Height to Diameter ratio	5.034
Height to top of Structure	2640.700 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Z_g = 274320.0$ mm.

Effective Height [z]:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 105.523 + 10000.0 = 10105.523$ mm.
 $= 33.155$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:
 Because z (33.155 ft.) > 15 ft.
 $= 2.01(z / Z_g)^{2/\alpha}$
 $= 2.01(33.155/900.0)^{2/9.5}$
 $= 1.003$

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
 $= 1$, per Table 26.6-1



As there is No Hill Present: [Kzt]:
 $K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:
 $= (1 + K1 \cdot K2 \cdot K3)^2$
 $= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$
 $= 1.0$

Velocity Pressure evaluated at height z, Imperial Units [qz]:
 $= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$
 $= \max(16, 0.00256 \cdot 1.003 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$
 $= 16.0 \text{ psf } [78.12] \text{ Kgs/m}^2$

Force on the first element [F]:
 $= qz \cdot G \cdot Cf \cdot \text{WindArea}$
 $= 16.0 \cdot 0.85 \cdot 0.567 \cdot 1.078$
 $= 8.3 \text{ lbs. } [3.8] \text{ Kgf}$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m2
bottom Dish	10105.5	0.000	0.000	0.000	1.003	1.000	78.120
Shell	11264.5	0.000	0.000	0.000	1.026	1.000	78.120
Shell Flange	12411.7	0.000	0.000	0.000	1.048	1.000	78.120
Dish flange	12523.7	0.000	0.000	0.000	1.049	1.000	78.120
Top dish	12661.7	0.000	0.000	0.000	1.052	1.000	78.120

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm.
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm.
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure mm.
 Lc - Total length of conical section(s) of vessel, mm.
 tb - Uncorroded plate thickness at bottom of vessel, mm.
 V30 - Design Wind Speed provided by user, Km/hr
 Vc - Critical wind velocity, Km/hr
 Vw - Maximum wind speed at top of structure, Km/hr
 W - Total corroded weight of structure, Kgf
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kgf
 Z - Maximum amplitude of vibration at top of vessel, mm.
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, $\leq 0.320\text{E-}06$ (High); $0.320\text{E-}06 < 0.400\text{E-}06$ (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: $0.0/2503.2 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[1.74/8.21^2] \cdot 10^4 = 258.088$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.400\text{E-}06$ no chance. [Vp]:
 $= W / (L \cdot Dr^2)$
 $= 1675 / (2503.2 \cdot 559.598^2)$
 $= 0.21369\text{E-}05$

Since Vp is > 0.400E-06 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping



ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf
Vortex breake	0.00	10000.00	78.12	0.00
FP	0.00	10000.00	78.12	0.00
Clip weight	0.00	10050.00	78.12	0.00
Demister	0.00	11650.00	78.12	0.00
demister supp	0.00	11650.00	78.12	0.00
Nozzles Linin	0.00	11050.00	78.12	0.00
Lug wt	0.00	10050.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 8.27802 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm^2	Wind Pressure Kgs/m^2	Element Wind Load Kgf
10	20	10105.5	633.6	1001.04	78.12	4.14755
20	30	11264.5	640.8	13943.8	78.12	57.7726
30	40	12411.7	607.2	680.064	78.12	2.81767
40	50	12523.7	607.2	680.064	78.12	2.81767
50	60	12661.7	633.6	1001.04	78.12	4.14755

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	1.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
= $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
= $0.020(13.53840.75)$
= 0.141 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
= (1/Natural Frequency) = (1/8.278)
= 0.121

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
= Minimum Value of ($1.7 \cdot 0.141$, 0.121) per 12.8.2
= 0.121

Loads on Building/Structure Supported Equipment:

Seismic Force per equation 13.3-1, [Fp]:
= $0.4 \cdot A_p \cdot Sds \cdot W(1 + 2 \cdot (z/h)) / (R / I_e)$
= $0.4 \cdot 1.0 \cdot 0.157 \cdot 1637(1 + 2 \cdot 1.0) / (2.0/1.25)$
= 192.463 Kgf

Maximum value of Fp per equation 13.3-2:
= $1.6 \cdot Sds \cdot I \cdot W$
= $1.6 \cdot 0.157 \cdot 1.25 \cdot 1637 = 513.23$ Kgf

Minimum value of Fp per equation 13.3-3:
= $0.3 \cdot 0.16 \cdot 1.25 \cdot 1637$
= 96.231 Kgf

Total Base Shear $V = F_p$, [V]:
= 192.463 Kgf

Vertical load per 12.4-4, [Ev]:
= $0.2 \cdot Sds \cdot W$
= $0.2 \cdot 0.157 \cdot 1637$
= 51.3 Kgf



Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x * h_{xk} / (\text{Sum of } W_i * h_{ik})$ and k is an exponent which is related
 to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 8.27802 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	25	160.08	0.34702	0.32601	-5.02011
20	Legs	300	121.259	3.15437	2.89644	-8.82278
Legs	30	1263	934.176	102.308	93.9427	42.5006
30	40	2285.2	161.252	31.9528	30.5742	13.2049
40	50	2397.2	161.252	33.5188	32.0727	8.14802
50	60	2478.2	98.5701	21.1817	19.1391	3.09116



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	331.977	...	...	...	...
20	Legs	125	4.14755	0.34702	0.34001	0.028448
Legs	30	963	71.7031	192.463	67.8113	279.092
30	40	1985.2	9.7829	86.6532	1.90025	13.9019
40	50	2097.2	6.96522	54.7005	0.96234	5.98592
50	60	2235.18	4.14755	21.1817	0.34001	1.73644

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kg	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	...	2.55423	2.55423	0.0021274
20	Legs	4.14755	2.55427	2.55439	0.0021287
Legs	30	71.7031	2.55754	2.56253	0.0021328
30	40	9.7829	2.56284	2.56314	0.0021328
40	50	6.96522	2.56345	2.56375	0.0021328
50	60	4.14755	2.56389	2.56402	0.0021328

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	94.1548	588.467
20	30	95.2247	595.155
30	40	90.2317	563.948
40	50	90.2317	563.948
50	60	94.1548	588.467

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
Allowable deflection : 12.516 Actual deflection : 2.564 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	178.662	162.734	2262127	2068267
20	30	228.708	212.78	2897351	2705670
30	40	228.708	212.78	2897351	2705670
40	50	178.662	162.734	2262127	2068267
50	60	178.662	162.734	2262127	2068267



Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	165.597	280.795	-142.49	-142.49
20	Legs	165.597	280.795	-145.736	-145.736
Legs	30	165.597	280.795	-145.736	-145.736
30	40	165.597	269.995	-145.736	-145.736
40	50	165.597	269.995	-142.49	-142.49
50	60	165.597	280.795	-142.49	-142.49

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	61.8873	...	80.4536
20	30	47.3771	...	61.5903
30	40	...	...	...
40	50	...	...	...
50	60	61.8874	...	80.4536

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	...	...	...	...	...
20	Legs	0.073764	0.07378	...	0.40747	0.40747
Legs	30	-0.62441	-0.62439	...	0.0011562	0.0011562
30	40	-0.19405	-0.19405	...	0.0011562	0.0011562
40	50	-0.15656	-0.15656	...	0.0015125	0.0015125
50	60	-0.059382	-0.059382	...	0.0015125	0.0015125

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	...	...	...	...	...
20	Legs	...	0.0012324	0.00010311	...	...
Legs	30	...	0.24578	1.01156	...	...
30	40	...	0.0068874	0.050387	...	...
40	50	...	0.0045629	0.028382	...	...
50	60	...	0.0016122	0.0082333	...	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	...
20	Legs	...	...	...	0.96868E-04
Legs	30	...	...	...	0.94504
30	40	...	...	...	0.046961
40	50	...	...	...	0.026119
50	60	...	...	...	0.0074393

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
------	----	--	---



10	20	...	-0.0030253
20	Legs	...	-0.0040663
Legs	30	...	0.019588
30	40	...	0.006086
40	50	...	0.0049102
50	60	...	0.0018628

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm^2	Earthquake For/Mom Corroded N./mm^2	Wind For/Mom No Corrosion N./mm^2	Earthquake For/Mom No Corrosion N./mm^2
10	20	...	...	...	...
20	Legs	...	...	...	...
Legs	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm^2
10	20	...
20	Legs	...
Legs	30	...
30	40	...
40	50	...
50	60	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
EP = Longitudinal Stress due to External Pressure
HP = Longitudinal Stress due to Hydrotest Pressure
NP = No Pressure
EW = Longitudinal Stress due to Weight (No Liquid)
OW = Longitudinal Stress due to Weight (Operating)
HW = Longitudinal Stress due to Weight (Hydrotest)
WI = Bending Stress due to Wind Moment (Operating)
EQ = Bending Stress due to Earthquake Moment (Operating)
EE = Bending Stress due to Earthquake Moment (Empty)
HI = Bending Stress due to Wind Moment (Hydrotest)
HE = Bending Stress due to Earthquake Moment (Hydrotest)
WE = Bending Stress due to Wind Moment (Empty) (no CA)
WF = Bending Stress due to Wind Moment (Filled) (no CA)
CW = Longitudinal Stress due to Weight (Empty) (no CA)
VO = Bending Stress due to Vortex Shedding Loads (Ope)
VE = Bending Stress due to Vortex Shedding Loads (Emp)
VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
FW = Axial Stress due to Vertical Forces for the Wind Case
FS = Axial Stress due to Vertical Forces for the Seismic Case
BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	142.49	0.0000	-0.0000
20	0.48	165.60	-0.33	145.74	0.0029	0.0023
20		165.60	-0.87	145.74		0.0060
50		165.60	-0.06	142.49		0.0004

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-0.00	142.49		0.0000
20	0.48	165.60	-0.34	145.74	0.0029	0.0023
20	0.34	165.60	-1.55	145.74	0.0021	0.0106
50		165.60	-0.07	142.49		0.0005

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	142.49	0.0000	-0.0000
20	0.48	165.60	-0.33	145.74	0.0029	0.0023
20		165.60	-0.87	145.74		0.0060
50		165.60	-0.06	142.49		0.0004



Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-0.00	142.49		0.0000
20	0.48	165.60	-0.34	145.74	0.0029	0.0023
20	0.41	165.60	-1.62	145.74	0.0025	0.0111
50		165.60	-0.07	142.49		0.0005

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	280.79	0.00	142.49	0.0000	-0.0000
20	0.00	280.79	0.00	145.74	0.0000	-0.0000
20	0.00	280.79	0.00	145.74	0.0000	-0.0000
50	0.00	280.79	0.00	142.49	0.0000	-0.0000

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	280.79	0.00	142.49	0.0000	-0.0000
20	0.00	280.79	0.00	145.74	0.0000	-0.0000
20	0.00	280.79	0.00	145.74	0.0000	-0.0000
50	0.00	280.79	0.00	142.49	0.0000	-0.0000

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	61.89	165.60		142.49	0.3737	
20	47.86	165.60		145.74	0.2890	
20	47.00	165.60		145.74	0.2838	
50	61.83	165.60		142.49	0.3734	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	61.88	165.60		142.49	0.3737	
20	47.85	165.60		145.74	0.2890	
20	47.79	165.60		145.74	0.2886	
50	61.84	165.60		142.49	0.3734	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	142.49	0.0000	-0.0000
20	0.48	165.60	-0.33	145.74	0.0029	0.0023
20		165.60	-0.87	145.74		0.0060
50		165.60	-0.06	142.49		0.0004

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		165.60	-0.00	142.49		0.0000
20	0.48	165.60	-0.34	145.74	0.0029	0.0023
20	0.41	165.60	-1.62	145.74	0.0025	0.0111
50		165.60	-0.07	142.49		0.0005

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	80.45	280.79		142.49	0.2865	
20	61.59	280.79		145.74	0.2193	



20	61.59	280.79	145.74	0.2193
50	80.45	280.79	142.49	0.2865

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	80.45	280.79		142.49	0.2865	
20	61.59	280.79		145.74	0.2193	
20	61.59	280.79		145.74	0.2193	
50	80.45	280.79		142.49	0.2865	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	61.89	165.60		142.49	0.3737	
20	47.86	165.60		145.74	0.2890	
20	46.75	165.60		145.74	0.2823	
50	61.83	165.60		142.49	0.3734	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	61.89	165.60		142.49	0.3737	
20	47.45	165.60		145.74	0.2865	
20	46.80	165.60		145.74	0.2826	
50	61.83	165.60		142.49	0.3734	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	61.89	165.60		142.49	0.3737	
20	47.86	165.60		145.74	0.2890	
20	46.75	165.60		145.74	0.2823	
50	61.83	165.60		142.49	0.3734	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	61.89	165.60		142.49	0.3737	
20	47.86	165.60		145.74	0.2890	
20	46.75	165.60		145.74	0.2823	
50	61.83	165.60		142.49	0.3734	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	165.60	0.00	142.49	0.0000	-0.0000
20	0.48	165.60	-0.33	145.74	0.0029	0.0023
20		165.60	-0.63	145.74		0.0043
50		165.60	-0.06	142.49		0.0004

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	61.88	138.00		118.74	0.4484	
20	47.85	138.00		121.45	0.3468	
20	46.77	138.00		121.45	0.3389	
50	61.83	138.00		118.74	0.4481	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10		138.00	-0.00	118.74		0.0000
20	0.48	138.00	-0.34	121.45	0.0035	0.0028
20		138.00	-0.61	121.45		0.0050
50		138.00	-0.06	118.74		0.0005

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	61.89	165.60		142.49	0.3737	
20	47.86	165.60		145.74	0.2890	
20	46.75	165.60		145.74	0.2823	
50	61.83	165.60		142.49	0.3734	

Absolute Maximum of the all of the Stress Ratio's 0.4484

Governing Element: bottom Dish

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
bottom Dish	61.9	...	...	...	61.9	+ Tens
bottom Dish	61.9	...	...	...	...	- Comp
Shell	47.4	0.1	0.4	...	47.9	+ Tens
Shell	47.4	0.1	-0.4	...	...	- Comp
Shell	47.4	-0.6	...	...	46.8	+ Tens
Shell	47.4	-0.6	...	...	...	- Comp
Top dish	61.9	-0.1	...	...	61.8	+ Tens
Top dish	61.9	-0.1	...	...	...	- Comp



Shop/Field Installation Options :

Lining is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Liquid	1023.651 mm.
Center of Gravity of the Lining	1293.156 mm.
Center of Gravity of the Nozzles	949.150 mm.
Center of Gravity of the Legs	-600.000 mm.
Center of Gravity of the Added Weights (Operating)	499.038 mm.
Center of Gravity of the Added Weights (Empty)	499.038 mm.
Center of Gravity of Bare Shell New and Cold	1607.279 mm.
Center of Gravity of Bare Shell Corroded	1624.852 mm.
Vessel CG in the Operating Condition	1145.424 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	1137.464 mm.
Vessel CG in the Test Condition	1166.443 mm.

Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	2500.00 mm.
Total vessel weight (No Liquid)	Twt 1712.97 Kg
Impact weight multiplication factor	Imp 1.50
Design lifting weight, DWT = Imp * Twt	2569.46 Kg
Elevation of the Tailing Lug (bottom)	150.00 mm.
Elevation of the Lifting Lug (top)	1690.00 mm.
Design Reaction force at the tailing lug	921.90 Kg
Design Reaction force at the lifting lug	1647.57 Kg

CG distance during field lifting operation	1137.46 mm.
CG distance from tailing lug	987.46 mm.
CG distance from the nearer lifting lug/trunnion	552.54 mm.

Compressive Allowable Increase Factor	1.00
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Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	3.50	1790.80	122.10 (UG-23)
Shear	0.79	1790.80	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kg	Shear Stress N./mm ²
0.00	0.0	0.0	452.4	0.2
1355.60	952.7	3.2	958.4	0.4
2270.80	117.1	0.4	1181.3	0.5
2360.40	48.6	0.2	246.7	0.1
2450.00	23.3	0.1	35.2	0.0

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.0287 at elevation 1790.7999 mm. - Must be <=1

Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ...----



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RESULTS FOR LEGS : Operating Case Description: LEGS

Legs attached to: Shell

Section Properties : Single Angle ISA80X80X10

India ISI Structural Steel Data

Overall Leg Length		1800.000	mm.
Effective Leg Length	Leglen	1500.000	mm.
Distance Leg Up Side of Vessel		250.000	mm.
Number of Legs	Nleg	3	
Cross Sectional Area for ISA80X80X10	Aleg	15.050	cm ²
Section Inertia (strong axis)		87.700	cm ⁴
Section Inertia (weak axis)		87.700	cm ⁴
Section Modulus (strong axis)		15500.063	mm. ³
Section Modulus (weak axis)		15500.063	mm. ³
Radius of Gyration (strong axis)		24.100	mm.
Radius of Gyration (weak axis)		24.100	mm.

Leg Orientation - Diagonal

Overturning Moment at top of Legs		279.1	Kg-m.
Total Weight Load at top of Legs	W	1692.9	Kgf
Total Shear force at top of Legs		192.5	Kgf
Additional force in Leg due to Bracing	Fadd	0.0	Kgf
Occasional Load Factor	Occfac	1.333	
Effective Leg End Condition Factor	k	1.000	

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Pad Width along Circumference	C11P	150.000	mm.
Pad Length along Vessel Axis	C22P	250.000	mm.
Pad Thickness	Tpad	16.000	mm.

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (192.5) (141.2/214.72)$
 $= 126.60 \text{ Kgf}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W/N_{\text{leg}} + (M_{\text{leg}} / (N_{\text{legm}} \cdot R_n))) / A_{\text{leg}}$
 $= (16602/3 + (2736740 / (1 \cdot 308.911))) / 1504.997$
 $= 9.56 \text{ N./mm}^2$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / N_{\text{leg}}) / A_{\text{leg}}$
 $= (1693/3) / 15.05$
 $= 3.68 \text{ N./mm}^2$

Computing Principal Axis and Inertias for Angle:

Leg lengths and thickness:	80.000	80.000	10.000
Distance to geometric centroid:	23.400	23.400	
Arm about YY:	18.400	21.600	
Arm about ZZ:	16.600	18.400	
Leg areas:	8.0000	7.0000	
Geometric inertia components YY:	27.751	61.242	
Geometric inertia components ZZ:	64.711	24.282	
Geometric inertias Iy & Iz:	88.993	88.993	
Product of inertia:	52.256		
Mohrs Radius:	52.256		
Average Inertia:	88.993		

QFACT = 1.0000 FBZ = 0.16383
 Principal Axis Inertias (Z&W) = 36.738 141.25



Angle to Principal Axis = 45.000
 Distances to extreme fibers CW & CZ = 56.568 23.476
 FOB from Eq 5-5 = 1.2986
 Bending allowables Fby & Fbz = 0.16383 0.16383

Shear Center Coordinates Wo & Zo: 25.095 0.0000

Values for Elastic Flexural-Torsional Buckling Stress:

E, G, J, R0²: 29500. 11346. 0.12053 2.8092
 AREA, LENGTH, Kw, Kz: 2.3328 59.055 1.0000 1.0000
 H, Few, Fez, Fej: 0.65252 121.45 31.588 208.68
 Fe computed from C4-1: 94.373

Initial (Kl/r)_{max}, & (Kl/r)_{equiv} = 96.007 55.544
 Final (Kl/r)_{max}, & Cc = 96.007 127.18
 Fa based on Eq 4-1 = 0.93617E-01

Leg Bending and Axial Stresses vs. Allowables: (N./mm²)

	Actual	Allowable
Weak Axis Bending :	84.15	218.38
Strong Axis Bending :	52.74	218.38
Axial Compression :	9.56	124.79

UNITY CHECKS ARE: H1-1 0.000
 H1-2 0.000
 H1-3 0.704

AISC Unity Check : 0.704, should be <= to 1 for Code compliance, [Passed]

WRC 107 Stress Analysis for Leg to Shell Junction, Ope Condition

Rectangular Attachment Parameter C11 113.137 mm.
 Rectangular Attachment Parameter C22 230.950 mm.

Input Echo, WRC107/537 Item 1, Description: LEGS

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Cylindrical	
Internal Corrosion Allowance	Cas	1.0000	mm.
Vessel Diameter	Dv	506.000	mm.
Vessel Thickness	Tv	14.000	mm.
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

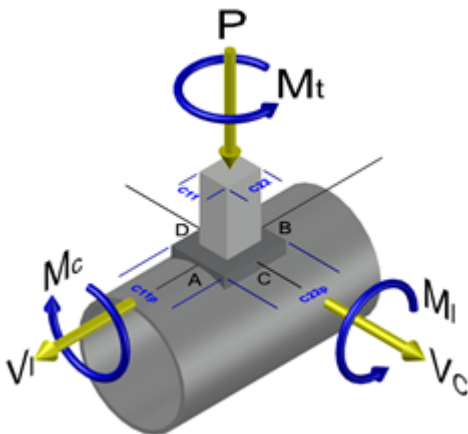
See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Rectangular	
Parameter C11	C11	113.14	mm.
Parameter C22	C22	230.95	mm.
Thickness of Reinforcing Pad	Tpad	16.000	mm.
Pad Parameter C11P	C11p	150.000	mm.
Pad Parameter C22P	C22p	250.000	mm.
Design Internal Pressure	Dp	49.507	bars
Include Pressure Thrust		No	
Vessel Centerline Direction Cosine	Vx	0.000	
Vessel Centerline Direction Cosine	Vy	1.000	
Vessel Centerline Direction Cosine	Vz	0.000	
Nozzle Centerline Direction Cosine	Nx	1.000	

Nozzle Centerline Direction Cosine	Ny	0.000	
Nozzle Centerline Direction Cosine	Nz	0.000	
Global Force (SUS)	Fx	0.0	Kgf
Global Force (SUS)	Fy	564.3	Kgf
Global Force (SUS)	Fz	0.0	Kgf
Global Moment (SUS)	Mx	0.0	Kg-m.
Global Moment (SUS)	My	0.0	Kg-m.
Global Moment (SUS)	Mz	23.7	Kg-m.
Internal Pressure (SUS)	P	49.51	bars
Include Pressure Thrust	No		
Global Force (OCC)	Fx	126.6	Kgf
Global Force (OCC)	Fy	903.5	Kgf
Global Force (OCC)	Fz	0.0	Kgf
Global Moment (OCC)	Mx	0.0	Kg-m.
Global Moment (OCC)	My	0.0	Kg-m.
Global Moment (OCC)	Mz	132.8	Kg-m.
Occasional Internal Pressure (OCC)	Pvar	0.00	bars

Use Interactive Control	No
WRC107 Version	Version March 1979
Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSstained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	564.3	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-23.7	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 9.26

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
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N(PHI) / (P/Rm)	0.399	4C	1.310	(A,B)
N(PHI) / (P/Rm)	0.399	3C	0.906	(C,D)
M(PHI) / (P)	0.276	2C1	0.061	(A,B)
M(PHI) / (P)	0.276	1C	0.084	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.267	3A !	0.321	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.294	1A	0.090	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.339	3B !	1.126	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.318	1B	0.032	(A,B,C,D)
N(x) / (P/Rm)	0.342	3C	1.031	(A,B)
N(x) / (P/Rm)	0.342	4C	1.389	(C,D)
M(x) / (P)	0.352	1C1	0.067	(A,B)
M(x) / (P)	0.352	2C	0.043	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.267	4A !	0.585	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.369	2A	0.045	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.339	4B !	0.370	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.381	2B	0.047	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.3	0.3	-0.3	-0.3	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		0.6	-0.6	-0.6	0.6	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		0.9	-0.3	-0.9	0.3	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.2	0.2	-0.2	-0.2	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		0.8	-0.8	-0.8	0.8	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		0.9	-0.6	-0.9	0.6	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Str. Int.		0.9	0.6	0.9	0.6	0.8	0.8	0.8	0.8

Dimensionless Parameters used : Gamma = 20.04

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.467	4C	1.991	(A,B)
N(PHI) / (P/Rm)	0.467	3C	1.004	(C,D)
M(PHI) / (P)	0.348	2C1	0.024	(A,B)
M(PHI) / (P)	0.348	1C !	0.057	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.341	3A	0.839	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.374	1A	0.073	(A,B,C,D)



N(PHI) / (ML/(Rm**2 * Beta))	0.405	3B	1.520	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.392	1B	0.016	(A,B,C,D)
N(x) / (P/Rm)	0.411	3C	1.189	(A,B)
N(x) / (P/Rm)	0.411	4C	2.201	(C,D)
M(x) / (P)	0.421	1C1	0.034	(A,B)
M(x) / (P)	0.421	2C !	0.030	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.341	4A	1.823	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.453	2A	0.030	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.405	4B	0.744	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.445	2B	0.022	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.9	0.9	-0.9	-0.9	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		1.3	-1.3	-1.3	1.3	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		2.2	-0.4	-2.2	0.4	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.6	0.6	-0.6	-0.6	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		1.6	-1.6	-1.6	1.6	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		2.2	-1.0	-2.2	1.0	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Str. Int.		2.2	1.0	2.2	1.0	1.7	1.7	1.7	1.7

WRC 107 Stress Calculation for OCCasional loads:

Radial Load	P	126.6	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	903.5	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-132.8	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 9.26

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.2	-0.2	-0.2	-0.2	-0.1	-0.1	-0.1	-0.1
Circ. Bend. P		-0.5	0.5	-0.5	0.5	-0.7	0.7	-0.7	0.7



Circ. Memb. MC	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC	-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML	1.8	1.8	-1.8	-1.8	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML	3.5	-3.5	-3.5	3.5	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.	4.5	-1.4	-6.0	2.0	-0.9	0.6	-0.9	0.6
Long. Memb. P	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
Long. Bend. P	-0.6	0.6	-0.6	0.6	-0.4	0.4	-0.4	0.4
Long. Memb. MC	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC	-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML	0.9	0.9	-0.9	-0.9	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	4.2	-4.2	-4.2	4.2	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	4.4	-2.9	-5.9	3.8	-0.6	0.2	-0.6	0.2
Shear VC	-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL	0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Shear MT	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear	0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Str. Int.	4.5	2.9	6.0	3.8	1.4	1.4	1.4	1.4

Dimensionless Parameters used : Gamma = 20.04
Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.7	-0.7	-0.7	-0.7	-0.4	-0.4	-0.4	-0.4
Circ. Bend. P		-1.0	1.0	-1.0	1.0	-2.5	2.5	-2.5	2.5
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		4.9	4.9	-4.9	-4.9	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		7.2	-7.2	-7.2	7.2	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		10.3	-2.1	-13.9	2.7	-2.9	2.2	-2.9	2.2
Long. Memb. P		-0.4	-0.4	-0.4	-0.4	-0.8	-0.8	-0.8	-0.8
Long. Bend. P		-1.5	1.5	-1.5	1.5	-1.3	1.3	-1.3	1.3
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		3.2	3.2	-3.2	-3.2	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		9.0	-9.0	-9.0	9.0	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		10.2	-4.7	-14.1	6.8	-2.1	0.5	-2.1	0.5
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-1.4	-1.4	1.4	1.4
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-1.4	-1.4	1.4	1.4
Str. Int.		10.3	4.7	14.1	6.8	3.9	3.2	3.9	3.2

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	41.0	46.0	41.0	46.0	41.0	46.0	41.0	46.0
Circ. Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pm(TOTAL)	41.0	46.0	41.0	46.0	41.0	46.0	41.0	46.0
Circ. Pl (SUS)	0.3	0.3	-0.3	-0.3	0.0	0.0	-0.0	-0.0
Circ. Pl (OCC)	1.6	1.6	-2.0	-2.0	-0.1	-0.1	-0.1	-0.1
Circ. Pl(TOTAL)	1.9	1.9	-2.3	-2.3	-0.1	-0.1	-0.1	-0.1
Circ. Q (SUS)	0.6	-0.6	-0.6	0.6	0.0	0.0	-0.0	0.0
Circ. Q (OCC)	2.9	-2.9	-4.0	4.0	-0.7	0.7	-0.7	0.7
Circ. Q (TOTAL)	3.6	-3.6	-4.6	4.6	-0.7	0.7	-0.7	0.7
Long. Pm (SUS)	20.5	20.5	20.5	20.5	20.5	20.5	20.5	20.5
Long. Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm(TOTAL)	20.5	20.5	20.5	20.5	20.5	20.5	20.5	20.5
Long. Pl (SUS)	0.2	0.2	-0.2	-0.2	0.0	0.0	-0.0	-0.0
Long. Pl (OCC)	0.7	0.7	-1.0	-1.0	-0.2	-0.2	-0.2	-0.2
Long. Pl(TOTAL)	0.9	0.9	-1.2	-1.2	-0.2	-0.2	-0.2	-0.2
Long. Q (SUS)	0.8	-0.8	-0.8	0.8	0.0	0.0	-0.0	0.0
Long. Q (OCC)	3.6	-3.6	-4.8	4.8	-0.4	0.4	-0.4	0.4
Long. Q (TOTAL)	4.4	-4.4	-5.6	5.6	-0.4	0.4	-0.4	0.4
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Shear Pl (OCC)	0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Shear Pl(TOTAL)	0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	41.0	46.0	41.0	46.0	41.0	46.0	41.0	46.0
Pm (SUS+OCC)	41.0	46.0	41.0	46.0	41.0	46.0	41.0	46.0
Pm+Pl (SUS)	41.3	46.3	40.7	45.7	41.0	46.0	41.0	46.0
Pm+Pl (SUS+OCC)	42.9	47.9	38.7	43.7	40.9	45.9	40.9	45.9
Pm+Pl+Q (Total)	46.5	44.3	34.1	48.3	40.2	46.6	40.2	46.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	45.97	138.00	Passed
Pm (SUS+OCC)	45.97	165.60	Passed
Pm+Pl (SUS)	46.29	207.00	Passed
Pm+Pl (SUS+OCC)	47.88	248.40	Passed
Pm+Pl+Q (TOTAL)	48.30	468.82	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		94.3	99.3	94.3	99.3	94.3	99.3	94.3	99.3
Circ. Pm (OCC)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pm(TOTAL)		94.3	99.3	94.3	99.3	94.3	99.3	94.3	99.3



Circ. P1 (SUS)	0.9	0.9	-0.9	-0.9	0.0	0.0	-0.0	-0.0
Circ. P1 (OCC)	4.1	4.1	-5.6	-5.6	-0.4	-0.4	-0.4	-0.4
Circ. P1(TOTAL)	5.0	5.0	-6.4	-6.4	-0.4	-0.4	-0.4	-0.4
Circ. Q (SUS)	1.3	-1.3	-1.3	1.3	0.0	0.0	-0.0	0.0
Circ. Q (OCC)	6.2	-6.2	-8.3	8.3	-2.5	2.5	-2.5	2.5
Circ. Q (TOTAL)	7.5	-7.5	-9.6	9.6	-2.5	2.5	-2.5	2.5
Long. Pm (SUS)	47.2	47.2	47.2	47.2	47.2	47.2	47.2	47.2
Long. Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm(TOTAL)	47.2	47.2	47.2	47.2	47.2	47.2	47.2	47.2
Long. P1 (SUS)	0.6	0.6	-0.6	-0.6	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)	2.7	2.7	-3.6	-3.6	-0.8	-0.8	-0.8	-0.8
Long. P1(TOTAL)	3.3	3.3	-4.2	-4.2	-0.8	-0.8	-0.8	-0.8
Long. Q (SUS)	1.6	-1.6	-1.6	1.6	0.0	0.0	-0.0	0.0
Long. Q (OCC)	7.5	-7.5	-10.5	10.5	-1.3	1.3	-1.3	1.3
Long. Q (TOTAL)	9.0	-9.0	-12.1	12.1	-1.3	1.3	-1.3	1.3
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-1.4	-1.4	1.4	1.4
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-2.2	-2.2	2.2	2.2
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	94.3	99.3	94.3	99.3	94.3	99.3	94.3	99.3
Pm (SUS+OCC)	94.3	99.3	94.3	99.3	94.3	99.3	94.3	99.3
Pm+P1 (SUS)	95.2	100.1	93.5	98.4	94.3	99.3	94.3	99.3
Pm+P1 (SUS+OCC)	99.3	104.3	87.9	92.8	94.1	99.0	94.1	99.0
Pm+P1+Q (Total)	106.8	96.8	78.3	102.4	91.5	101.5	91.5	101.5

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	99.27	138.00	Passed
Pm (SUS+OCC)	99.27	165.60	Passed
Pm+P1 (SUS)	100.14	207.00	Passed
Pm+P1 (SUS+OCC)	104.26	248.40	Passed
Pm+P1+Q (TOTAL)	106.79	468.82	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

Bolting Size Requirement for Leg Baseplates :

Baseplate Material	SA-516 70
Baseplate Allowable Stress	SBA 137.90 N./mm2
Baseplate Length	B 154.0000 mm.
Baseplate Width	D 154.0000 mm.
Baseplate Thickness	BTHK 16.0000 mm.
Leg Dimension Along Baseplate Length	d 79.9998 mm.
Leg Dimension Along Baseplate Width	b 79.9998 mm.
Bolt Circle Diameter	BCD 484.0000 mm.
Bolt Material	SA-307 B
Bolt Allowable Tensile Stress	STBA 138.00 N./mm2
Bolt Allowable Shear Stress	SBShear 82.80 N./mm2
Anchor Bolt Nominal Diameter	BOD 24.0000 mm.
Number of Anchor Bolts in Tension per Leg	NB 0
Total Number of Anchors Bolt per Leg	NBT 1



Bolt Corrosion Allowance	bca	3.0000 mm.
Ultimate 28-day Concrete Strength	FCPRIME	50.000 N./mm2

Leg Baseplate and Bolting Analysis, including Moments

Shear Stress in a Single Bolt [tb]:
= Shear Force/(Bolt Area • Number of Bolts)
= 192.5/(1.53 • 3)
= 4.1 N./mm². Must be less than 82.8 N./mm².

Plate thickness required (5/8"; 16mm) per D. Moss

Since there is only 1 bolt, the standard computation for this geometry is not possible. A reasonable bolt size should be selected. Per Megyesy, use a 1 inch (25.4 mm) bolt.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: D1-2"

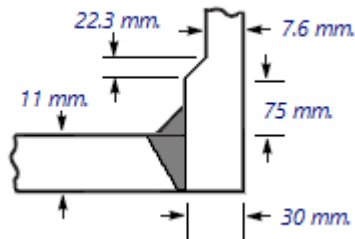
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Pressure for Reinforcement Calculations	P	3.660	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516	70
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	506.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	11.0000	mm.
Head Internal Corrosion Allowance	cas	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	22.3000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm.
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	11.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		500.0	mm.
Centerline Distance to Tangent		500.0	mm.

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: D1-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.773 in.
Actual Thickness Used in Calculation	0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.66 \cdot 508.0 \cdot 0.8966 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 3.66)$
 $= 0.6043$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.66 \cdot 23.52 / (138 \cdot 1.0 - 0.6 \cdot 3.66)$
 $= 0.0625$ mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 29.0$, less than the hub height, use (e-2)
 $= 72.5000$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	125.0342	mm.
Parallel to Vessel Wall	Rn+tn+t	62.5171	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	25.0000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: D1-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.0625	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 1.6704	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 2.5000	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 4.4200	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.42, \max(2.5, 2.5)]$
 $= 2.5000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(1.0625, 2.5)$
 $= 2.5000$ mm.

Available Nozzle Neck Thickness = 7.6454 mm. -> OK



Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	40.3,	Allowable	:	138.0 N./mm2	Passed
Expansion	:	0.0,	Allowable	:	304.7 N./mm2	Passed
Occasional	:	0.6,	Allowable	:	183.5 N./mm2	Passed
Shear	:	28.8,	Allowable	:	96.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 7.645, tr = 0.062, c = 1.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.009, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 11.0, tr = 0.604, c = 1.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.06, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 3.66/102.00 = 0.036

Weld Size Calculations, Description: D1-2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 2.9803 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 213.9803 mm.

Input Echo, WRC107/537 Item 1, Description: D1-2" :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Spherical
Internal Corrosion Allowance	Cas	1.0000 mm.
Vessel Diameter	Dv	910.800 mm.
Vessel Thickness	Tv	11.000 mm.
Design Temperature	T1	130.0 C
Vessel Material		SA-516 70
Vessel UNS Number		K02700
Vessel Cold S.I. Allowable	Smc	138.00 N./mm2
Vessel Hot S.I. Allowable	Smh	138.00 N./mm2



See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	ID
Corrosion Allowance for Nozzle	Can	1.0000 mm.
Nozzle Diameter	Dn	45.034 mm.
Nozzle Thickness	Tn	30.000 mm.
Nozzle Material		SA-105
Nozzle UNS Number		K03504
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²
Design Internal Pressure	Dp	3.660 bars
Include Pressure Thrust		No

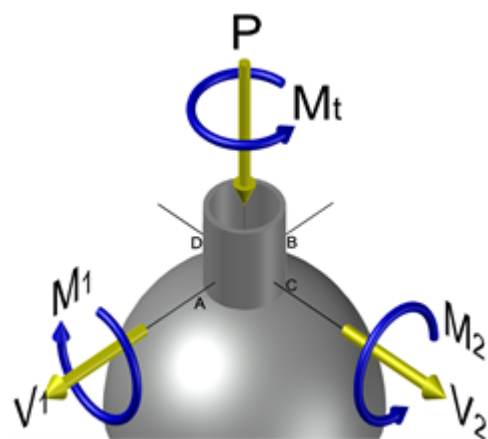
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	382.4	Kgf
Longitudinal Shear (SUS)	(V1) V1	254.9	Kgf
Circumferential Shear (SUS)	(Vc) V2	254.9	Kgf
Circumferential Moment (SUS)	(Mc) M1	35.7	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	35.7	Kg-m.
Torsional Moment (SUS)	Mt	51.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 105.034 + 2 \cdot 1.65 \cdot \sqrt{(461.4 (11.0 - 1.0))}$$

$$= 329.191 \text{ mm.}$$

WRC 107 Stress Calculation for SUSstained loads:

Radial Load	P	382.4	Kgf
Circumferential Shear	(VC) V2	254.9	Kgf
Longitudinal Shear	(VL) V1	254.9	Kgf

Circumferential Moment	(MC)	M1	35.7	Kg-m.
Longitudinal Moment	(ML)	M2	35.7	Kg-m.
Torsional Moment		MT	51.0	Kg-m.

Dimensionless Param: U = 0.77 TAU = 5.00 (1.31) RHO = 0.34

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.05242	(A,B,C,D)
$M(x) / P$	SP 1	0.05954	(A,B,C,D)
$N(x) * T * \text{SQRT}(Rm * T) / MC$	SM 1	0.06361	(A,B,C,D)
$M(x) * \text{SQRT}(Rm * T) / MC$	SM 1	0.12533	(A,B,C,D)
$N(x) * T * \text{SQRT}(Rm * T) / ML$	SM 1	0.06361	(A,B,C,D)
$M(x) * \text{SQRT}(Rm * T) / ML$	SM 1	0.12533	(A,B,C,D)
$N(y) * T / P$	SP 1	0.08137	(A,B,C,D)
$M(y) / P$	SP 1	0.01694	(A,B,C,D)
$N(y) * T * \text{SQRT}(Rm * T) / MC$	SM 1	0.08213	(A,B,C,D)
$M(y) * \text{SQRT}(Rm * T) / MC$	SM 1	0.03967	(A,B,C,D)
$N(y) * T * \text{SQRT}(Rm * T) / ML$	SM 1	0.08213	(A,B,C,D)
$M(y) * \text{SQRT}(Rm * T) / ML$	SM 1	0.03967	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0
Rad.	Bend. P	-13.4	13.4	-13.4	13.4	-13.4	13.4	-13.4	13.4
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-3.3	-3.3	3.3	3.3
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-38.7	38.7	38.7	-38.7
Rad.	Memb. ML	-3.3	-3.3	3.3	3.3	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-38.7	38.7	38.7	-38.7	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-57.4	46.9	26.6	-24.0	-57.4	46.9	26.6	-24.0
Tang. Memb. P		-3.1	-3.1	-3.1	-3.1	-3.1	-3.1	-3.1	-3.1
Tang.	Bend. P	-3.8	3.8	-3.8	3.8	-3.8	3.8	-3.8	3.8
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-4.2	-4.2	4.2	4.2
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-12.3	12.3	12.3	-12.3
Tang.	Memb. ML	-4.2	-4.2	4.2	4.2	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-12.3	12.3	12.3	-12.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-23.4	8.8	9.6	-7.3	-23.4	8.8	9.6	-7.3
Shear VC		1.5	1.5	-1.5	-1.5	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.5	-1.5	1.5	1.5
Shear MT		2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9
Tot. Shear		4.4	4.4	1.4	1.4	1.4	1.4	4.4	4.4
Str. Int.		57.9	47.4	26.8	24.1	57.4	46.9	27.7	25.1

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm2)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	8.4	8.4	8.4	8.4	8.4	8.4	8.4	8.4



Rad. P1 (SUS)	-5.2	-5.2	1.3	1.3	-5.2	-5.2	1.3	1.3
Rad. Q (SUS)	-52.1	52.1	25.3	-25.3	-52.1	52.1	25.3	-25.3
Long. Pm (SUS)	8.4	8.4	8.4	8.4	8.4	8.4	8.4	8.4
Long. P1 (SUS)	-7.3	-7.3	1.2	1.2	-7.3	-7.3	1.2	1.2
Long. Q (SUS)	-16.1	16.1	8.4	-8.4	-16.1	16.1	8.4	-8.4
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	1.5	1.5	-1.5	-1.5	-1.5	-1.5	1.5	1.5
Shear Q (SUS)	2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9
Pm (SUS)	8.4	8.4	8.4	8.4	8.4	8.4	8.4	8.4
Pm+P1 (SUS)	3.9	3.9	11.1	11.1	3.9	3.9	11.1	11.1
Pm+P1+Q (Total)	49.6	55.7	35.1	17.0	49.1	55.3	36.1	18.9

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.36	138.00	Passed
Pm+P1 (SUS)	11.12	207.00	Passed
Pm+P1+Q (TOTAL)	55.75	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N1-3"

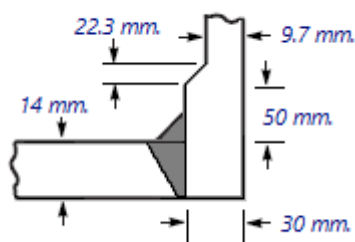
From: 20

Pressure for Reinforcement Calculations	P	3.660	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	506.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1750.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	50.0000	mm.
Height of Beveled Transition	L'	22.3300	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm.
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	11.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N1-3"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 3.500 in.
Actual Thickness Used in Calculation 0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $3.66 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.66)$
= 0.6748 mm.

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.3468 and the shell thk. is 14.0000 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $3.66 \cdot 44.45 / (138 \cdot 1.0 + 0.4 \cdot 3.66)$
= 0.1178 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 20.265 / 0.5773
= 35.1008 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	142.8618	mm.
Parallel to Vessel Wall, opening length	d	71.4309	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N1-3".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.1178	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 1.6748	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 2.5000	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 5.8000	mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[5.8, max(2.5, 2.5)]
= 2.5000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.1178, 2.5)
= 2.5000 mm.

Available Nozzle Neck Thickness = 9.7345 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):



Sustained :	29.7,	Allowable :	138.0 N./mm2	Passed
Expansion :	0.0,	Allowable :	315.3 N./mm2	Passed
Occasional :	0.7,	Allowable :	183.5 N./mm2	Passed
Shear :	24.1,	Allowable :	96.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 9.735$, $t_r = 0.118$, $c = 1.0$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.013$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 14.0$, $t_r = 0.675$, $c = 1.0$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.052$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.66 / 102.00 = 0.036$

Weld Size Calculations, Description: N1-3"

Intermediate Calc. for nozzle/shell Welds T_{min} 13.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.9353 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 267.9353 mm.

Input Echo, WRC297 Item 1, Description: N1-3" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	506.000	mm.
Shell Thickness	14.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm2
Vessel Hot S.I. Allowable	Smh	138.00 N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD
Nozzle Corrosion Allowance	1.0000 mm.
Nozzle Diameter	129.431 mm.



Nozzle Thickness		30.0000	mm.
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	3.66	bars
Radial Load	P	-612.00	Kgf
Circumferential Shear	Vc	408.00	Kgf
Longitudinal Shear	Vl	408.00	Kgf
Circumferential Moment	Mc	92.00	Kg-m.
Longitudinal Moment	Ml	92.00	Kg-m.
Torsional Moment	Mt	131.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (4.314) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 129.431 + 2 \cdot 1.65 \cdot \sqrt{(260.5(14.0 - 1.0))}$
 $= 321.470 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	521.000	mm.
Nozzle Outside Diameter	(d) =	129.431	mm.
Vessel Thickness used	(T) =	13.000	mm.
Nozzle Thickness used	(t) =	29.000	mm.
T / t	=	0.448	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.573$$

Nr/P	=	0.079
Mr/P	=	0.168
M0/P	=	0.049
N0/P	=	0.106
MrD/Mc	=	0.295
NrDL/Mc	=	0.096
M0d/Mc	=	0.092
N0DL/Mc	=	0.061
MrD/Ml	=	0.138
NrDL/Ml	=	0.079
M0D/Ml	=	0.044
N0DL/Ml	=	0.127

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5



Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	6	7	6	7
Outplane Stress Summary	4	6	37	-5

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	35	-35	35	-35
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-34	34	34	-34
Inplane Pressure Stress	3	3	3	3
Inplane Stress Summary	4	1	79	-60

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	5	5	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	10	11	79	60

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-22	22	22	-22
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	3	3	3	3
Outplane Stress Summary	-7	16	42	-23

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	35	-35	35	-35
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-73	73	73	-73
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	6	7	6	7
Inplane Stress Summary	-31	43	122	-94

Vessel Stresses (N./mm²):



CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)				
	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)				
	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	31	43	122	95

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)				
	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	0	0	0
Hoop Stress Summary	-1	0	9	9

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)				
	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-4	-4	4	4
Axial Bending (ML)	-4	4	4	-4
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-3	-4	15	-4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)				
	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	2	2	1	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)				
	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	5	6	16	14

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)				
	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3



Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	0	0	0

Hoop Stress Summary	1	2	6	7
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-4	-4	4	4
Axial Bending (Mc)	-12	12	12	-12
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-10	2	23	-12
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	2	2
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	12	3	23	19

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	7	6	7	6	7	6	7
Circ. Pl (SUS)	-2	-2	8	8	-1	-1	5	5
Circ. Q (SUS)	0	0	20	-20	-38	38	108	-108
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-1	-1	5	5	1	1	5	5
Long. Q (SUS)	1	-1	69	-69	-12	12	32	-32
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	6.0	7.0	6.0	7.0	6.0	7.0	6.0	7.0
Pm+Pl (SUS)	4.4	5.3	14.2	15.1	5.6	6.4	11.3	12.2
Pm+Pl+Q (Total)	8.1	8.9	77.1	61.1	33.2	44.1	119.2	96.2

Vessel Stress Summation Comparison (N./mm²):

Type of	Max. S.I.	S.I. Allowable	Result
---------	-----------	----------------	--------



Stress Int.			
Pm (SUS)	7.00	138.00	Passed
Pm+Pl (SUS)	15.14	207.00	Passed
Pm+Pl+Q (TOTAL)	119.20	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm2):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	0	0	0	0	0	0	0	0
Circ. Pl (SUS)	-2	-2	8	8	1	1	5	5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-4	-4	4	4	-4	-4	4	4
Long. Q (SUS)	1	-1	9	-9	-7	7	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)	4.0	4.0	8.0	8.0	5.0	5.0	5.0	5.0
Pm+Pl+Q (Total)	3.6	5.3	13.2	13.2	12.2	3.4	21.1	18.1

Nozzle Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	8.00	207.00	Passed
Pm+Pl+Q (TOTAL)	21.06	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: M1-12"

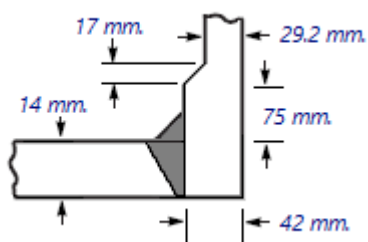
From: 20

Pressure for Reinforcement Calculations	P	3.660	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	506.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		800.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	17.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		42.0000	mm.
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).





ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 12.750 in.
Actual Thickness Used in Calculation 1.148 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $3.66 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.66)$
= 0.6748 mm.

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.3468 and the shell thk. is 14.0000 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $3.66 \cdot 161.925 / (138 \cdot 1.0 + 0.4 \cdot 3.66)$
= 0.4291 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 12.841/0.5773
= 22.2409 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

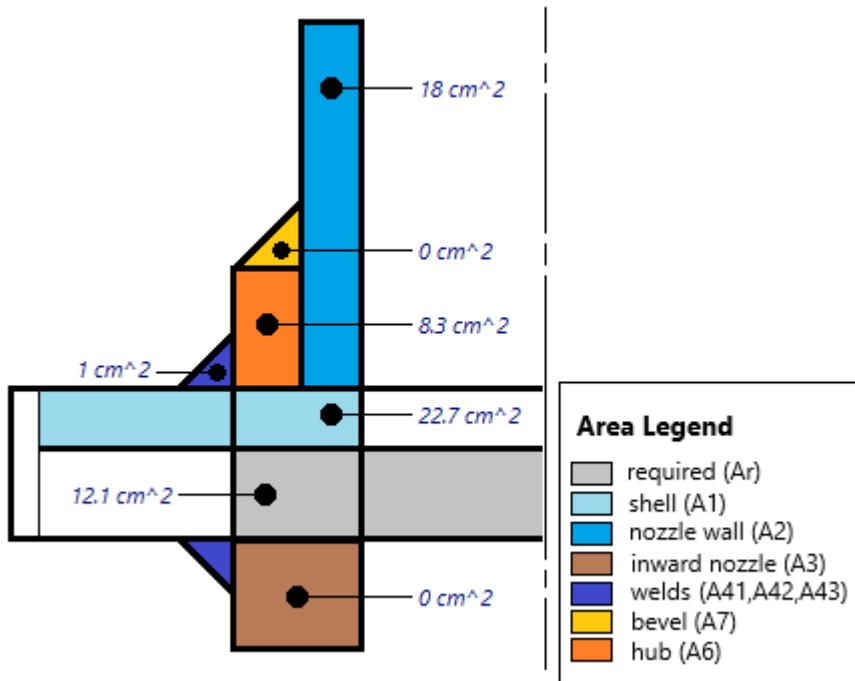
UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	535.0632	mm.
Parallel to Vessel Wall, opening length	d	267.5316	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm.

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	12.061	NA	NA
Area in Shell	A1	22.719	NA	NA
Area in Nozzle Wall	A2	18.025	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	8.347	NA	NA
TOTAL AREA AVAILABLE	Atot	50.090	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \cdot UG-37(c)$$

$$= (267.5316 \cdot 4.5081 \cdot 1.0 + 2 \cdot 41.0 \cdot 4.5081 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 12.061 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 267.532 (1.0 \cdot 13.0 - 1.0 \cdot 4.508) - 2 \cdot 41.0$$

$$(1.0 \cdot 13.0 - 1.0 \cdot 4.5081) \cdot (1 - 1.0)$$

$$= 22.719 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 32.5) (28.16 - 0.43) 1.0$$

$$= 18.025 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$
 $= (2 \cdot \min(32.5, 200.0, 75.0)) \cdot (42.0 - 29.2) \cdot 1.0$
 $= 8.347 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 1.4291 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= t_a$
 $= 1.4291 \text{ mm}$.

Available Nozzle Neck Thickness = 29.1592 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 29.159$, $t_r = 0.429$, $c = 1.0 \text{ mm}$., $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.015$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 14.0$, $t_r = 0.675$, $c = 1.0 \text{ mm}$., $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.052$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.66/102.00 = 0.036$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 12.841/0.5773$
 $= 22.2409 \text{ mm}$.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 401.2974 mm.
 Parallel to Vessel Wall, opening length d 200.6487 mm.
 Normal to Vessel Wall (Thickness Limit), no pad T_{lnp} 32.5000 mm.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	8.041	8.041
Area Available in Shell	A1	11.359	11.359
Area Available in Nozzle Wall	A2	18.025	18.025
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	1.000	1.000



Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	8.347	8.347
TOTAL AREA AVAILABLE	Atot	38.730	38.730

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: M1-12"

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (12.0606 - 22.7185 + 2 \cdot 41.0 \cdot 1.0 \cdot (1.0 \cdot 13.0 - 4.5081)) \cdot 138)$
 $= \max(0, -5198.85) \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (26.3711 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 38515.46 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (26.3711 + 0.0 + 1. + (10.66)) \cdot 138$
 $= 53515.76 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (26.3711 + 0.0 + 1. + 0.0 + (10.66)) \cdot 138$
 $= 53515.76 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 323.85 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 35075. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 147.8454) \cdot (42.0 - 1.0) \cdot 0.7 \cdot 138$
 $= 187578. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 323.85 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 63565. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (35075 + 187578) = 222653 \text{ Kgf}$



Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (35075 + 0 + 63565 + 0) = 98641 Kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (35075 + 63565 + 0) = 98641 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 222653 Kgf, must exceed W = 0 Kgf	or W1 = 38515 Kgf
Path 2-2 = 98640 Kgf, must exceed W = 0 Kgf	or W2 = 53515 Kgf
Path 3-3 = 98640 Kgf, must exceed W = 0 Kgf	or W3 = 53515 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 58.6053 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 272.6053 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 267.53 mm.

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 71.93 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 267.53 mm.

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Input, Nozzle Desc: K3-2"

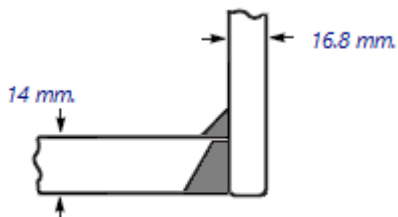
From: 20

Pressure for Reinforcement Calculations	P	3.660	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	506.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1000.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K3-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----



Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.66 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.66)$
 $= 0.6748$ mm.

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.3468 and the shell thk. is 14.0000 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([3.66 / (138 \cdot 1.0)] - 1))$
 $= 0.0701$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	110.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	55.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.0701	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 1.6748	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 2.5000	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 5.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(2.5, 2.5)]$
 $= 2.5000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.0701, 2.5)$
 $= 2.5000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 14.0, tr = 0.675, c = 1.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.052$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $3.66/102.00 = 0.036$

Weld Size Calculations, Description: K3-2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.3882 mm.

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Input, Nozzle Desc: K1-2"

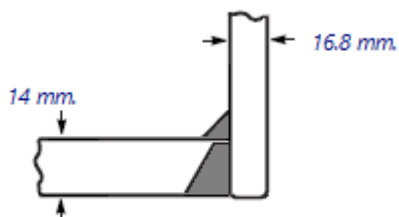
From: 20

Pressure for Reinforcement Calculations	P	3.660	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	506.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1000.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K1-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----



Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.66 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.66)$
 $= 0.6748$ mm.

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.3468 and the shell thk. is 14.0000 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \left(\exp \left(\frac{P}{S \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
 $= 26.4 \left(\exp \left(\frac{3.66}{138 \cdot 1.0} \right) - 1 \right)$
 $= 0.0701$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	110.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	55.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.0701	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 1.6748	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 2.5000	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 5.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(2.5, 2.5)]$
 $= 2.5000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.0701, 2.5)$
 $= 2.5000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 14.0, tr = 0.675, c = 1.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.052$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $3.66/102.00 = 0.036$

Weld Size Calculations, Description: K1-2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.3882 mm.

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Input, Nozzle Desc: K2-2"

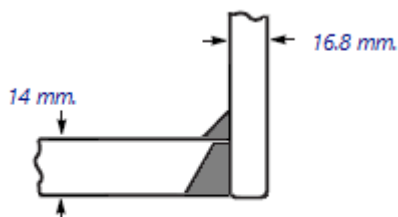
From: 20

Pressure for Reinforcement Calculations	P	3.660	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	506.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		300.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K2-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
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Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $3.66 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.66)$
= 0.6748 mm.

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.3468 and the shell thk. is 14.0000 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
= $26.4 \left(\exp \left(\frac{3.66}{138.0 \cdot 1.0} \right) - 1 \right)$
= 0.0701 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	110.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	55.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2-2".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.0701	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 1.6748	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 2.5000	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 5.8000	mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[5.8, max(2.5, 2.5)]
= 2.5000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.0701, 2.5)
= 2.5000 mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 14.0, tr = 0.675, c = 1.0 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.052$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $3.66/102.00 = 0.036$

Weld Size Calculations, Description: K2-2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.3882 mm.

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Input, Nozzle Desc: K4-2"

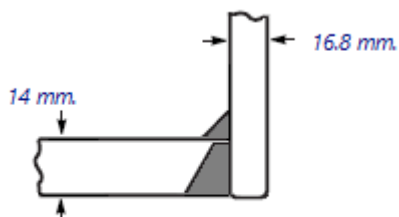
From: 20

Pressure for Reinforcement Calculations	P	3.660	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	506.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		300.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K4-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
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Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $3.66 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.66)$
= 0.6748 mm.

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.3468 and the shell thk. is 14.0000 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
= $26.4 \left(\exp \left(\frac{3.66}{138.0 \cdot 1.0} \right) - 1 \right)$
= 0.0701 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	110.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	55.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K4-2".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.0701	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 1.6748	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 2.5000	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 5.8000	mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[5.8, max(2.5, 2.5)]
= 2.5000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.0701, 2.5)
= 2.5000 mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 14.0, tr = 0.675, c = 1.0 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.052$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $3.66/102.00 = 0.036$

Weld Size Calculations, Description: K4-2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.3882 mm.

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Input, Nozzle Desc: N2-3"

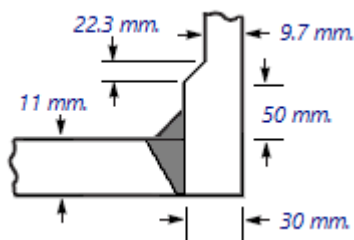
From: 50

Pressure for Reinforcement Calculations	P	3.660	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	506.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	11.0000	mm.
Head Internal Corrosion Allowance	cas	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	100.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	50.0000	mm.
Height of Beveled Transition	L'	22.3300	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm.
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	11.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.



Reinforcement CALCULATION, Description: N2-3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.66 \cdot 508.0 \cdot 0.8966 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 3.66)$
 $= 0.6042$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.66 \cdot 44.45 / (138 \cdot 1.0 + 0.4 \cdot 3.66)$
 $= 0.1178$ mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 20.265 / 0.5773$
 $= 35.1008$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	146.3771	mm.
Parallel to Vessel Wall, opening length	d	73.1886	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	25.0000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N2-3".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govern. thk, tg = 9.735, tr = 0.118, c = 1.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.013$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govern. thk, tg = 11.0, tr = 0.604, c = 1.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.06$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $3.66/102.00 = 0.036$

Weld Size Calculations, Description: N2-3"

Intermediate Calc. for nozzle/shell Welds T_{min} 13.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N2-3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $3.66 \cdot 508.0 \cdot 0.8966 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 3.66)$
= 0.6042 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $3.66 \cdot 44.45 / (138 \cdot 1.0 + 0.4 \cdot 3.66)$
= 0.1178 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= $20.265 / 0.5773$
= 35.1008 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	142.8618 mm.
Parallel to Vessel Wall, opening length	d	71.4309 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	25.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N2-3".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.1178 mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 1.6704 mm.
Wall Thickness	tb1	= max(trb1, tr16b) = 2.5000 mm.



Wall Thickness $tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 5.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(2.5, 2.5)]$
 $= 2.5000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.1178, 2.5)$
 $= 2.5000 \text{ mm.}$

Available Nozzle Neck Thickness = 9.7345 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	29.7,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	315.3 N./mm ²	Passed
Occasional	:	0.7,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	24.1,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: N2-3"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 13.0000 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 12.1480 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 273.4061 mm.

Warning:
The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N2-3" :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Spherical
Internal Corrosion Allowance	Cas	1.0000 mm.
Vessel Diameter	Dv	910.800 mm.
Vessel Thickness	Tv	11.000 mm.
Design Temperature	T1	130.0 C
Vessel Material		SA-516 70
Vessel UNS Number		K02700
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow

Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	1.0000	mm.
Nozzle Diameter	Dn	129.431	mm.
Nozzle Thickness	Tn	30.000	mm.
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm2
Design Internal Pressure	Dp	3.660	bars
Include Pressure Thrust		No	

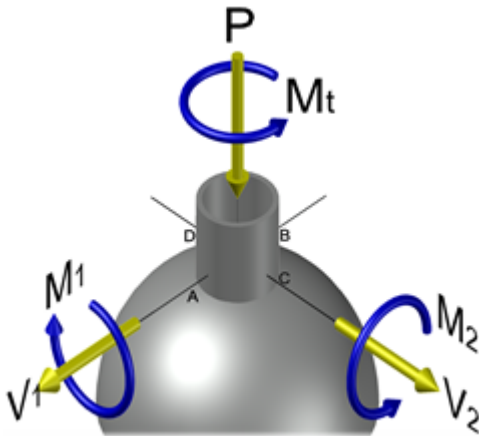
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-612.0	Kgf
Longitudinal Shear (SUS)	(V1) V1	408.0	Kgf
Circumferential Shear (SUS)	(Vc) V2	408.0	Kgf
Circumferential Moment (SUS)	(Mc) M1	92.0	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	92.0	Kg-m.
Torsional Moment (SUS)	Mt	131.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 129.431 + 2 \cdot 1.65 \cdot \sqrt{(461.4 (11.0 - 1.0))}$$

$$= 353.588 \text{ mm.}$$

WRC 107 Stress Calculation for SUS tained loads:

Radial Load	P	-612.0	Kgf
Circumferential Shear	(VC) V2	408.0	Kgf
Longitudinal Shear	(VL) V1	408.0	Kgf
Circumferential Moment	(MC) M1	92.0	Kg-m.
Longitudinal Moment	(ML) M2	92.0	Kg-m.
Torsional Moment	MT	131.0	Kg-m.

Dimensionless Param: U = 0.95 TAU = 5.00 (1.73) RHO = 0.34



Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 1	0.04305	(A,B,C,D)
M(x) / P	SP 1	0.04668	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 1	0.04960	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 1	0.08886	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 1	0.04960	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 1	0.08886	(A,B,C,D)
N(y) * T / P	SP 1	0.06615	(A,B,C,D)
M(y) / P	SP 1	0.01318	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 1	0.06652	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 1	0.02707	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 1	0.06652	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 1	0.02707	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
Rad. Bend. P		16.8	-16.8	16.8	-16.8	16.8	-16.8	16.8	-16.8
Rad. Memb. MC		0.0	0.0	0.0	0.0	-6.6	-6.6	6.6	6.6
Rad. Bend. MC		0.0	0.0	0.0	0.0	-70.8	70.8	70.8	-70.8
Rad. Memb. ML		-6.6	-6.6	6.6	6.6	0.0	0.0	0.0	0.0
Rad. Bend. ML		-70.8	70.8	70.8	-70.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-58.0	50.0	96.8	-78.5	-58.0	50.0	96.8	-78.5
Tang. Memb. P		4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
Tang. Bend. P		4.7	-4.7	4.7	-4.7	4.7	-4.7	4.7	-4.7
Tang. Memb. MC		0.0	0.0	0.0	0.0	-8.8	-8.8	8.8	8.8
Tang. Bend. MC		0.0	0.0	0.0	0.0	-21.6	21.6	21.6	-21.6
Tang. Memb. ML		-8.8	-8.8	8.8	8.8	0.0	0.0	0.0	0.0
Tang. Bend. ML		-21.6	21.6	21.6	-21.6	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-21.7	12.0	39.1	-13.5	-21.7	12.0	39.1	-13.5
Shear VC		2.0	2.0	-2.0	-2.0	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-2.0	-2.0	2.0	2.0
Shear MT		4.9	4.9	4.9	4.9	4.9	4.9	4.9	4.9
Tot. Shear		6.8	6.8	2.9	2.9	2.9	2.9	6.8	6.8
Str. Int.		59.3	51.2	96.9	78.6	58.2	50.2	97.6	79.2

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		8.4	8.4	8.4	8.4	8.4	8.4	8.4	8.4
Rad. Pl (SUS)		-4.0	-4.0	9.2	9.2	-4.0	-4.0	9.2	9.2
Rad. Q (SUS)		-54.0	54.0	87.6	-87.6	-54.0	54.0	87.6	-87.6
Long. Pm (SUS)		8.4	8.4	8.4	8.4	8.4	8.4	8.4	8.4
Long. Pl (SUS)		-4.9	-4.9	12.8	12.8	-4.9	-4.9	12.8	12.8
Long. Q (SUS)		-16.8	16.8	26.3	-26.3	-16.8	16.8	26.3	-26.3



Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	2.0	2.0	-2.0	-2.0	-2.0	-2.0	2.0	2.0
Shear Q (SUS)	4.9	4.9	4.9	4.9	4.9	4.9	4.9	4.9
Pm (SUS)	8.4	8.4	8.4	8.4	8.4	8.4	8.4	8.4
Pm+P1 (SUS)	5.9	5.9	22.0	22.0	5.9	5.9	22.0	22.0
Pm+P1+Q (Total)	50.9	59.6	105.3	70.2	49.9	58.6	106.0	70.8

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.36	138.00	Passed
Pm+P1 (SUS)	22.02	207.00	Passed
Pm+P1+Q (TOTAL)	105.96	413.99	Passed

*Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: K11-2"

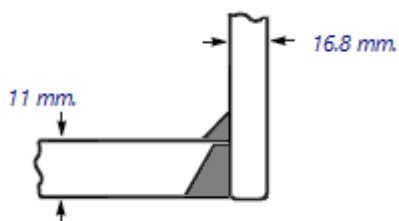
From: 50

Pressure for Reinforcement Calculations	P	3.660	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	506.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	11.0000	mm.
Head Internal Corrosion Allowance	cas	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	100.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		225.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	11.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K11-2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.66 \cdot 508.0 \cdot 0.8966 / (2 \cdot 138.0 \cdot 1.0 \cdot 0.2 \cdot 3.66)$
 $= 0.6042$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)]) - 1)$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([3.66 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0701$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	108.1069	mm.
Parallel to Vessel Wall, opening length	d	54.0534	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	25.0000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K11-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 11.0, tr = 0.604, c = 1.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.06$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.66 / 102.00 = 0.036$

Weld Size Calculations, Description: K11-2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.



Reinforcement CALCULATION, Description: K11-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.66 \cdot 508.0 \cdot 0.8966 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 3.66)$
 $= 0.6042$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S \cdot E)] - 1) - 1)$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([3.66 / (138.0 \cdot 1.0)] - 1) - 1)$
 $= 0.0701$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.6000	mm.
Parallel to Vessel Wall, opening length	d	52.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	25.0000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K11-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.0701	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 1.6704	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 2.5000	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 5.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(2.5, 2.5)]$
 $= 2.5000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.0701, 2.5)$
 $= 2.5000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: K11-2"

Intermediate Calc. for nozzle/shell welds Tmin 13.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 11.4060 mm.

KOD for H2 Compressor (175-D)
PV Elite 28 U1 Licensee: SL User
FileName : 175-D 7%
Nozzle Calcs.: K11-2"



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Nozl: 29 2:33pm Sep 16, 2026



HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.5142 mm.

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Input, Nozzle Desc: K12-2"

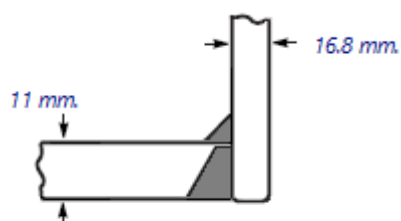
From: 50

Pressure for Reinforcement Calculations	P	3.660	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	506.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	11.0000	mm.
Head Internal Corrosion Allowance	cas	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	100.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		350.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	11.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K12-2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.66 \cdot 508.0 \cdot 0.8966 / (2 \cdot 138.0 \cdot 1.0 \cdot 0.2 \cdot 3.66)$
 $= 0.6042$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)]) - 1)$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([3.66 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0701$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	108.1069	mm.
Parallel to Vessel Wall, opening length	d	54.0534	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	25.0000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K12-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 11.0, tr = 0.604, c = 1.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.06$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.66 / 102.00 = 0.036$

Weld Size Calculations, Description: K12-2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.



Reinforcement CALCULATION, Description: K12-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.66 \cdot 508.0 \cdot 0.8966 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 3.66)$
 $= 0.6042$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot \exp([3.66 / (138.0 \cdot 1.0)] - 1)$
 $= 0.0701$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	105.6000	mm.
Parallel to Vessel Wall, opening length	d	52.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	25.0000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K12-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.0701	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 1.6704	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 2.5000	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 5.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(2.5, 2.5)]$
 $= 2.5000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.0701, 2.5)$
 $= 2.5000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: K12-2"

Intermediate Calc. for nozzle/shell welds	Tmin	13.0000	mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 11.4060 mm.

KOD for H2 Compressor (175-D)
PV Elite 28 U1 Licensee: SL User
FileName : 175-D 7%
Nozzle Calcs.: K12-2"



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The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.5142 mm.

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: